

Lesson 1: Introduction to SAS Visual Text Analytics



1.0 Lesson Overview



Lesson Overview

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1.1 SAS Visual Text Analytics



Overview of Text Analytics



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SAS® Visual Text Analytics



- Sepsis indicators
- Painkiller addict
- Environmental influences



- Is product meeting quality expectations?



- Terrorism
- Cyber bullying
- At-risk persons



- What amenities influence bookings?

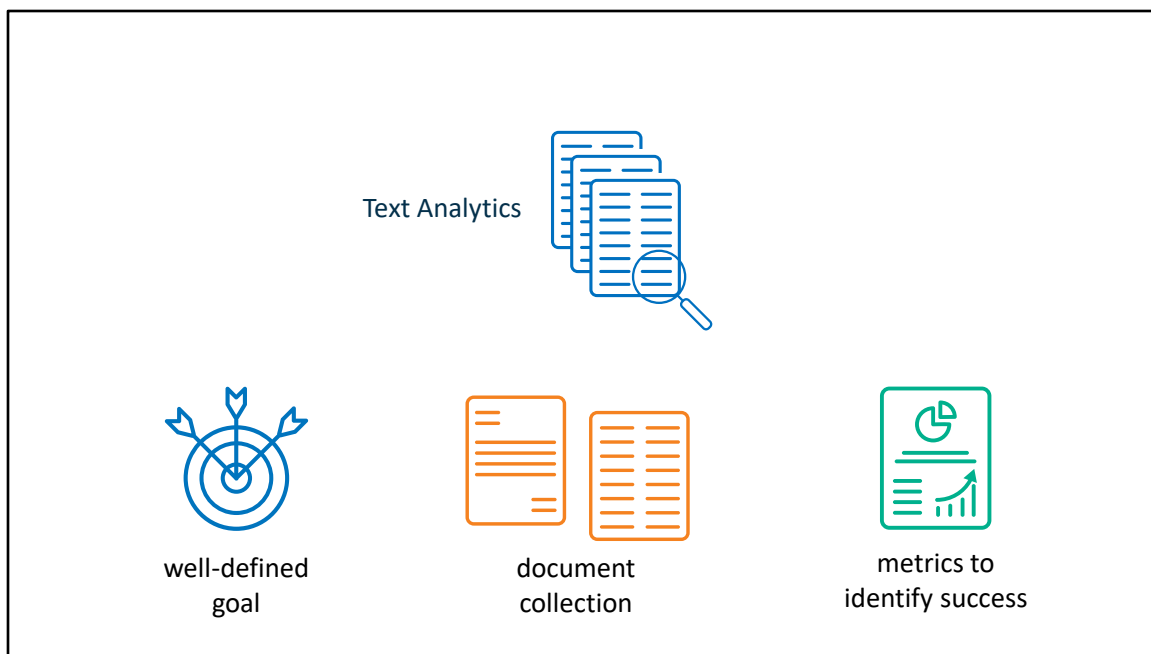
Why is SAS Visual Text Analytics important? SAS Visual Text Analytics automatically converts unstructured data into meaningful insights. Considering the staggering amount of unstructured data that's generated every day, automation is critical to fully analyze text and speech data efficiently.

Sepsis is the leading cause of death in hospitals, and it must be detected early or it can be life threatening. Sepsis is an inflammatory response to infection. The toxins from bacteria in the body release into the bloodstream to fight the threat, causing systemic inflammation that can slow blood flow and damage organs. If sepsis progresses to the later stages of septic shock, it can be life threatening. If sepsis is properly treated in a timely manner, we've seen that we can save up to 80 percent of the lives that we're losing. Taking advantage of data from the electronic medical records of all patients in its facilities, the solution uses natural language processing and a rules engine to continually monitor factors that could indicate a sepsis infection.

When it comes to customer reviews, one could simply report in terms of a numerical metric "Average Customer Review: 4 stars". However, when customer comments are analyzed, or patterns in those comments are extracted, a better picture of the product is obtained.

Consider social media like Facebook and Twitter. Unstructured data has a great amount of information. If text patterns are extracted and series of behaviors detected, then interventions for cyber bullying, sexual predatory behavior, and terrorism planning could be done timely.

Think about web sites like TripAdvisor. Reviews from Airbnb from specific geographic areas can help determine how important amenities are to securing bookings and customer satisfaction.

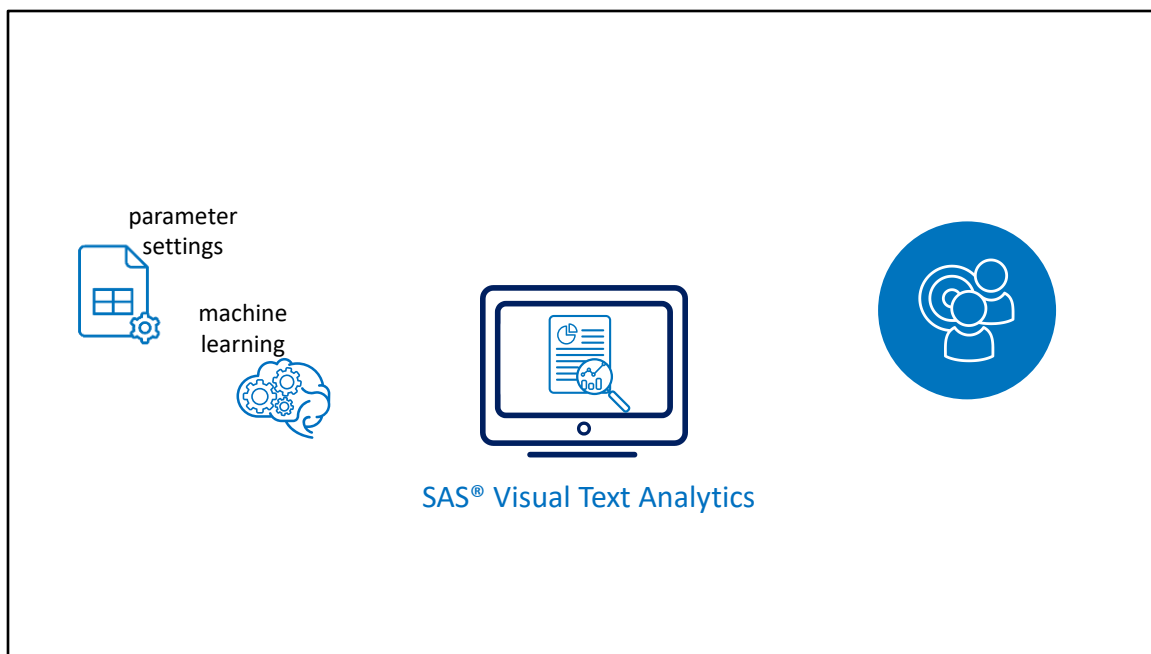


Text analytics project requirements include a well-defined goal, document collection, and metrics to identify success.

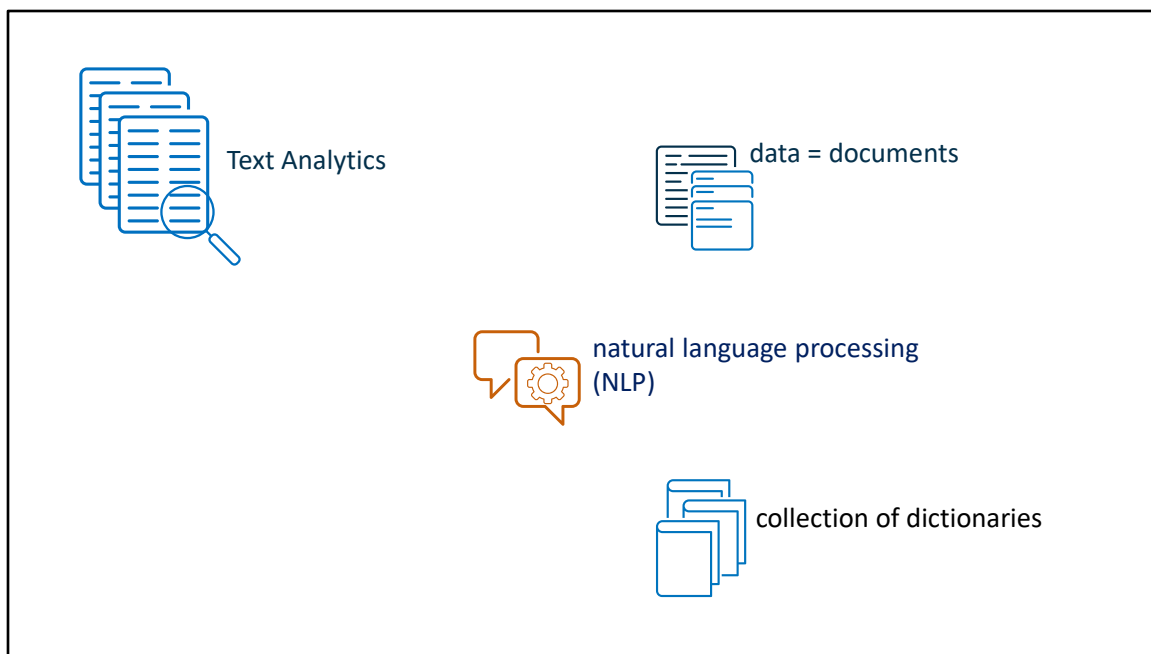


SAS® Visual Text Analytics

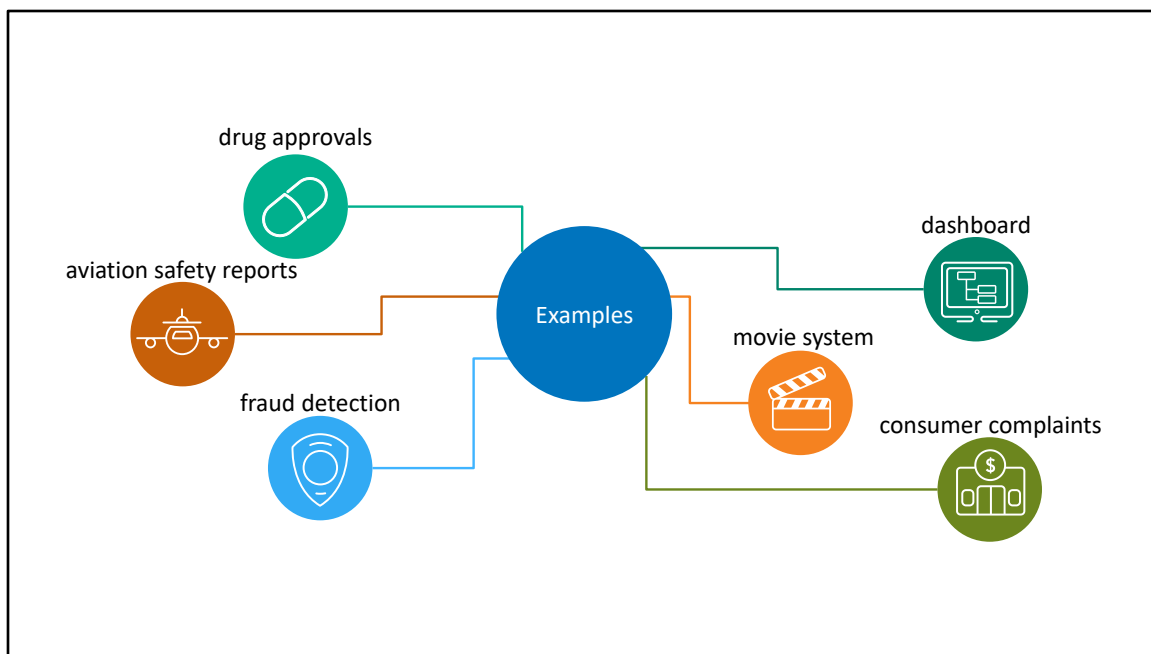
SAS Visual Text Analytics is a high-end software product that simplifies the use of text analytics for many different types of users, from business analysts to document librarians.



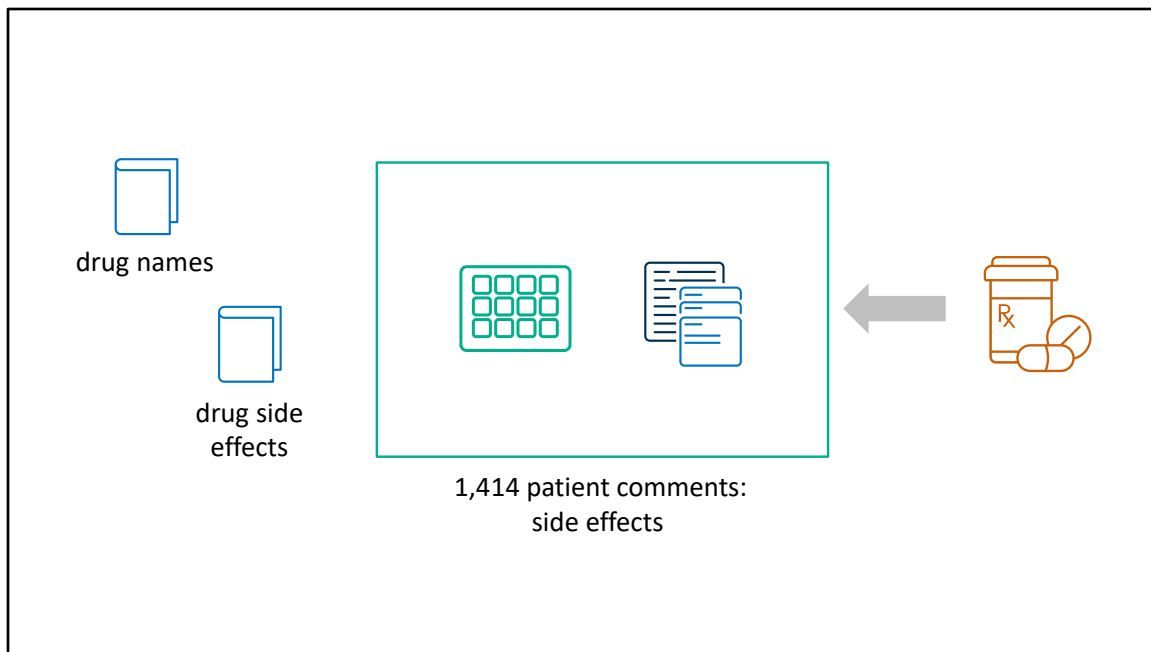
By choosing intelligent parameter settings, or by using machine-learning tools to “learn” an appropriate setting, Visual Text Analytics frees the user to concentrate on the immediate task.



The principal analysis object in text analytics is a document. Data for a text analytics project consists of a collection of documents. A feature of SAS Visual Text Analytics is the ability to convert document collections stored in a variety of forms into a form that can be processed using the software. A key algorithmic component of text analytics is natural language processing. In addition, projects are aided by dictionaries that encapsulate domain expertise. They exploit human knowledge that is related to the documents. A key to success is software that can rapidly and consistently process, analyze, and score documents.



Many examples of successful text analytics projects exist. In regulatory filings for drug approvals, you can find all dosage amounts for medications that are mentioned in the application. In order to promote customer satisfaction, you can populate a dashboard with the top 10 customer complaints for the week. To expedite processing aviation safety reports, you can identify all reports that involve a runway incursion. You can enhance the accuracy of a movie recommendation system by exploiting the relationships between movie synopses and customer movie ratings. You can enhance the accuracy of a fraud and abuse detection system for workers' compensation insurance by exploiting the relationships between adjuster notes, policy-based model inputs, claim-based model inputs, and a fraud target variable. You could rapidly and consistently identify all consumer complaints that are related to home loans.



In our first demonstration, we're going to consider a text analytics project that seeks to extract drug dosages from a collection of customer feedback documents. The data set contains 1,414 patient comments about drug side effects. The documents are in the English language. Two dictionaries are available, one that contains drug names, and another that contains a drug's side effects. The software employed to analyze the documents is SAS Visual Text Analytics.

Demo: Extracting Drug Dosages from User-Feedback Documents, Part 1

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Demo: Extracting Drug Dosages from User-Feedback Documents, Part 2

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Demo: Extracting Drug Dosages from User-Feedback Documents, Part 3

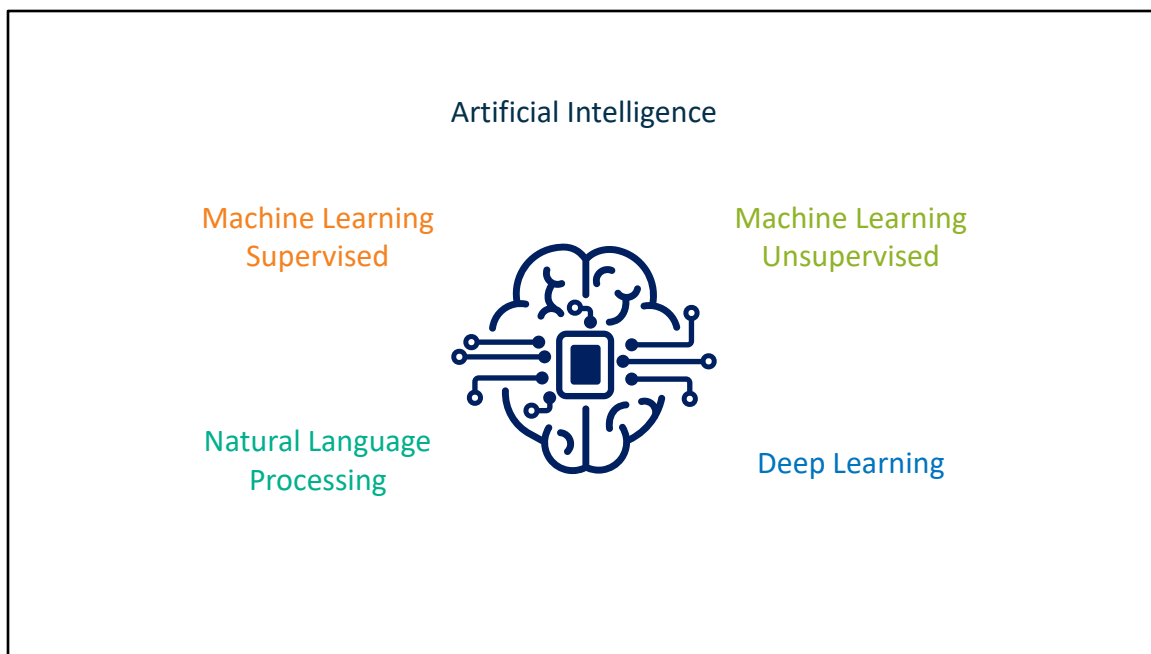
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Overview of Natural Language Processing

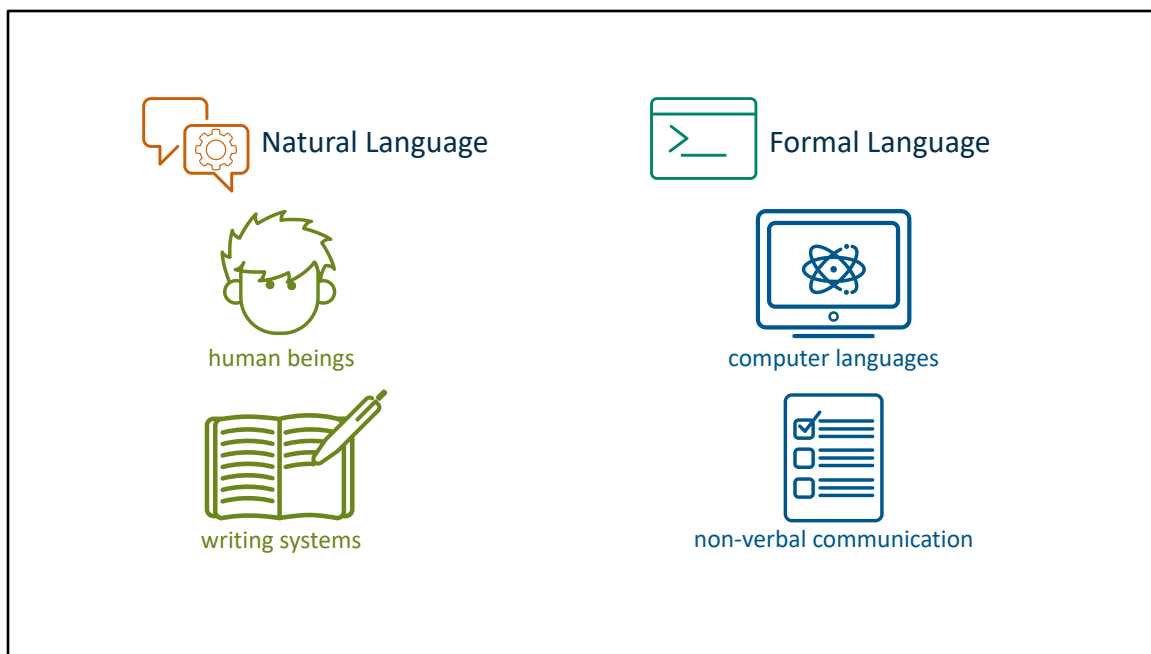


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Artificial Intelligence is a scientific discipline that explores how human intelligence functions, with the goal of replicating human intelligence using machines. There are many subfields of artificial intelligence, including machine learning, natural language processing, and deep learning. Natural language processing, abbreviated NLP, benefits from research in computer science, computational linguistics, and other areas to bridge the gap between human communication and computer understanding.

While supervised and unsupervised learning, and specifically deep learning, are now widely used for modeling human language, there's also a need for syntactic and semantic understanding and domain expertise that are not necessarily present in these machine learning approaches.



Natural language processing is the key algorithmic component to SAS Visual Text Analytics. *Natural language* refers to verbal language spoken by human beings, for example, English, Spanish, French, and so on. Most human languages have been encoded using a writing system. *Formal language* refers to C, C++, C#, Java, and other computer languages, or any man-made language designed for any purpose other than verbal communication between human beings.

A formal language has rigid syntactical rules to avoid ambiguities. Some references use the term *artificial language* rather than formal language. However, artificial language can also refer to languages constructed for purposes other than communication.

Artificial Intelligence



Natural Language Processing (NLP)



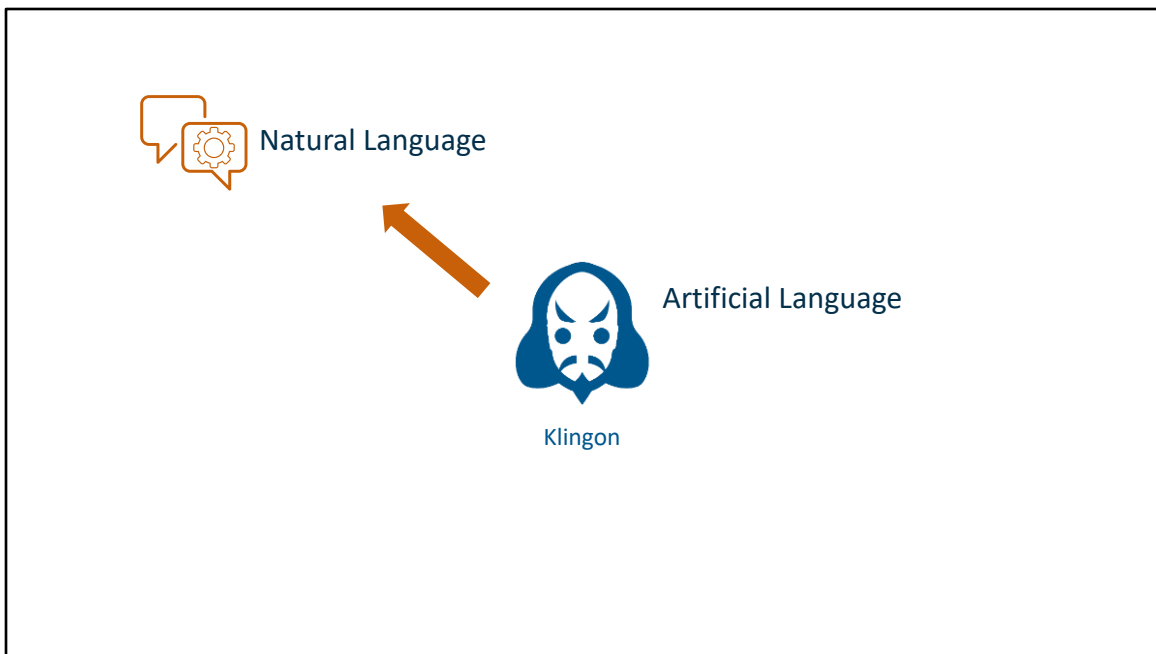
natural language



specific purposes



Natural language processing uses computers to process natural language for specific purposes. For example, one specific purpose is to identify components of a language, such as parts-of-speech in the English language.



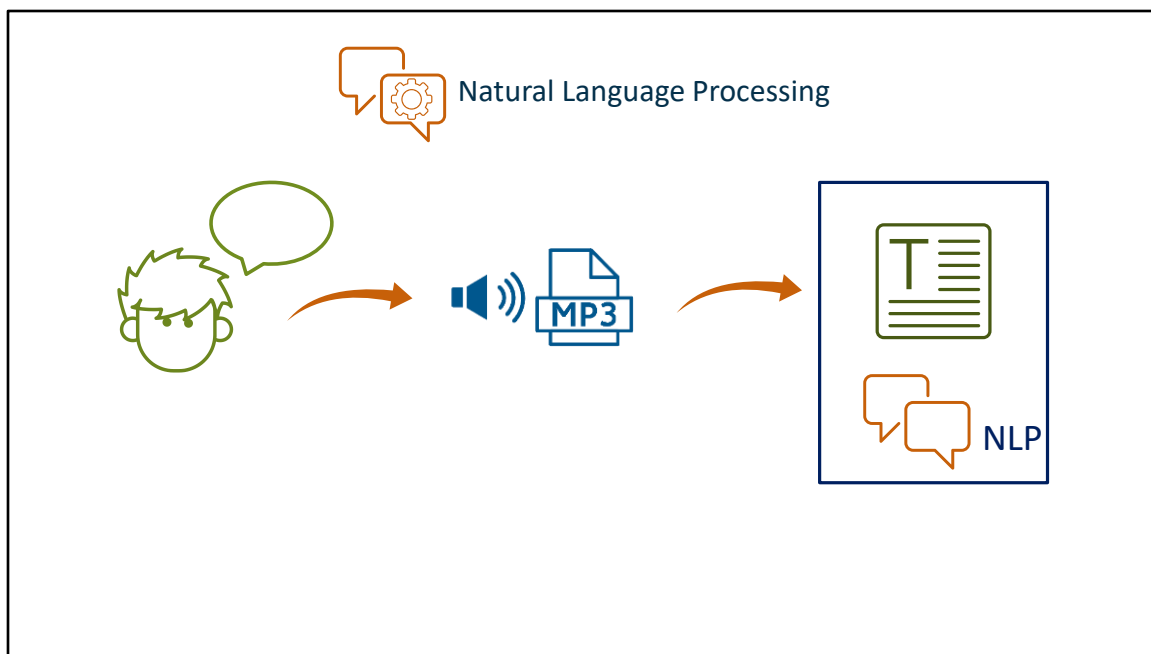
There are invented languages like Klingon developed for the Star Trek science fiction television and movie series. Such languages are often called artificial languages, but in our taxonomy, an artificial language would be treated just like a natural language.



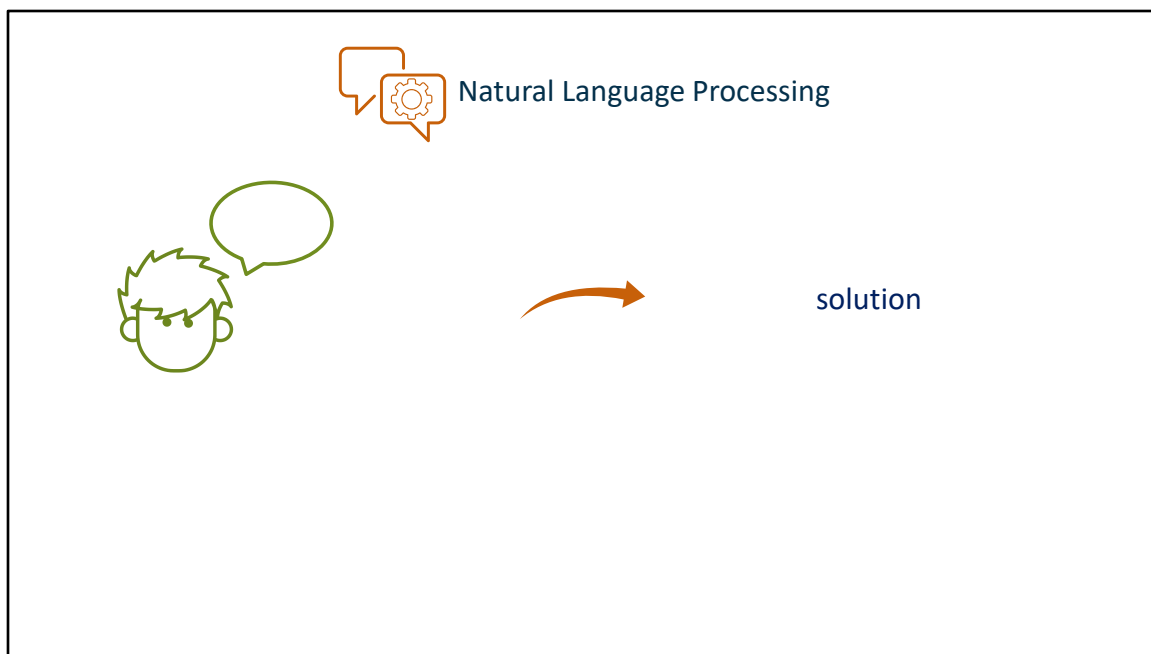
Natural Language Processing



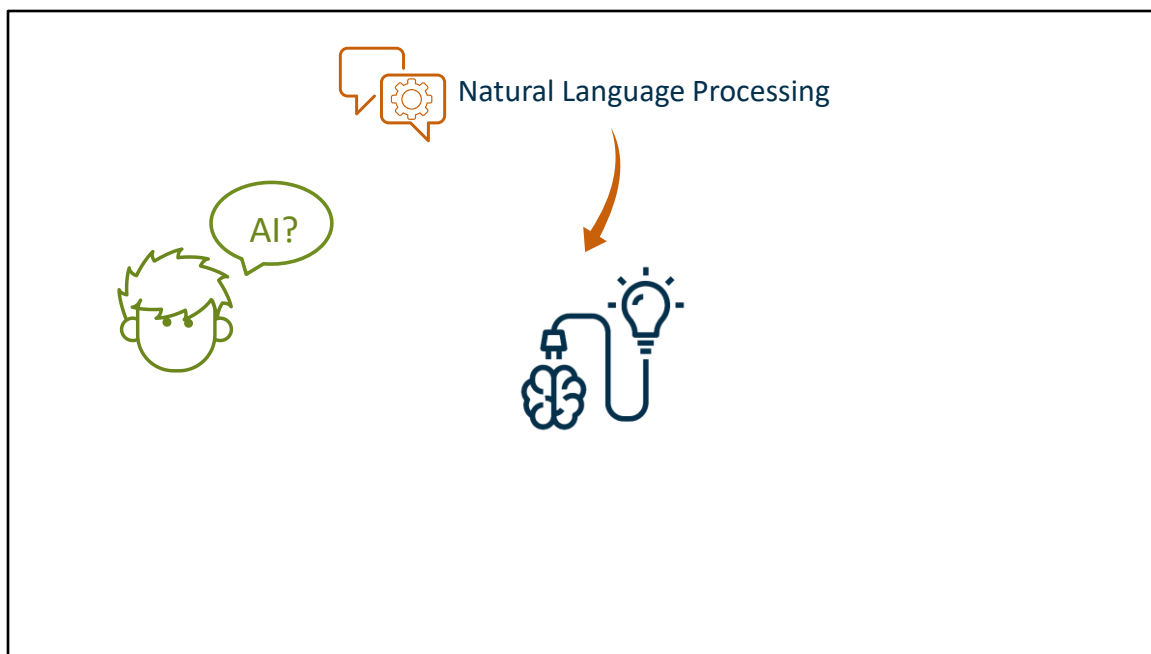
The goal of NLP is to process both spoken and written language.



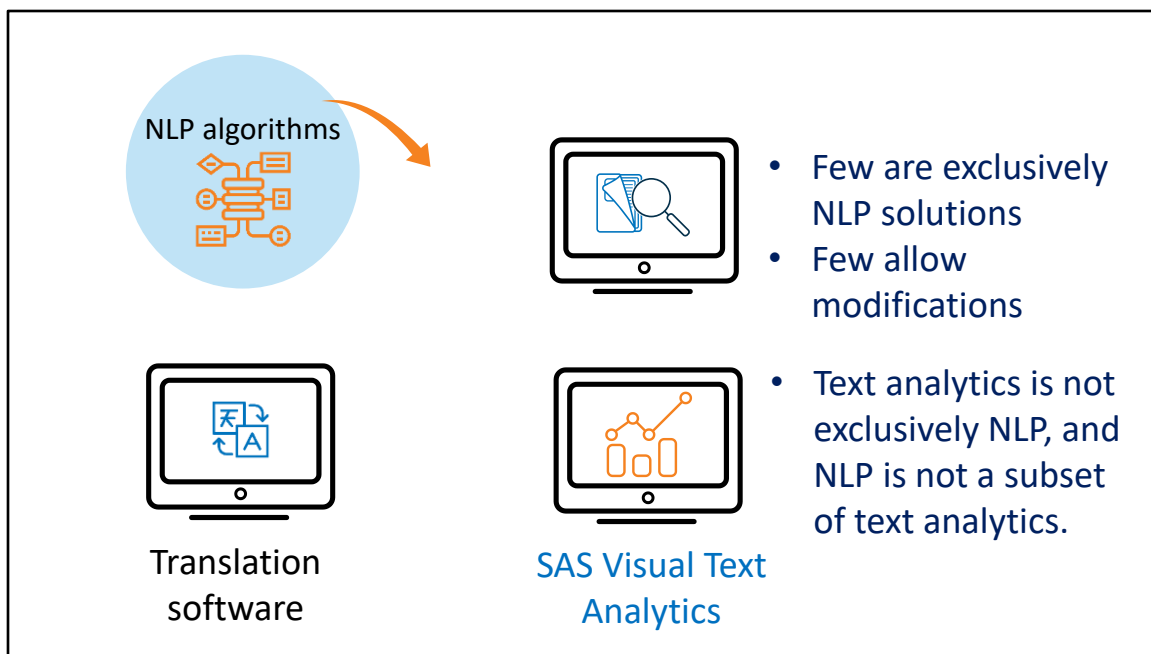
If you think about the process of going from spoken language to text analytics, the first step is to convert spoken language to digitized audio. For example, you could record a conversation as an MP3 file. The second step is to convert the digitized audio of the spoken language to written language. Speech recognition software performs this conversion. The last step is to process the written language using Natural Language Processing.



Some complete NLP solutions, like Apple Siri and Amazon Alexa, proceed directly from spoken language to solution.



One goal of SAS Visual Text Analytics is to unleash the power of natural language processing without burdening the user with prerequisite knowledge of artificial intelligence.



Text analytics software includes integrated NLP algorithms. NLP algorithms are also used in text processing software not intended for analytical use, such as translation software. Many advanced software products use NLP, and some of these would not be classified as text analytics software products.

Few software products are exclusively NLP solutions. Many software products that use NLP do not permit users to modify hyperparameters that are employed by the NLP algorithms.

SAS Visual Text Analytics and other SAS text analytics products use NLP algorithms embedded within the software. Text analytics is not exclusively NLP, and NLP is not a subset of text analytics.

Natural Language Processing

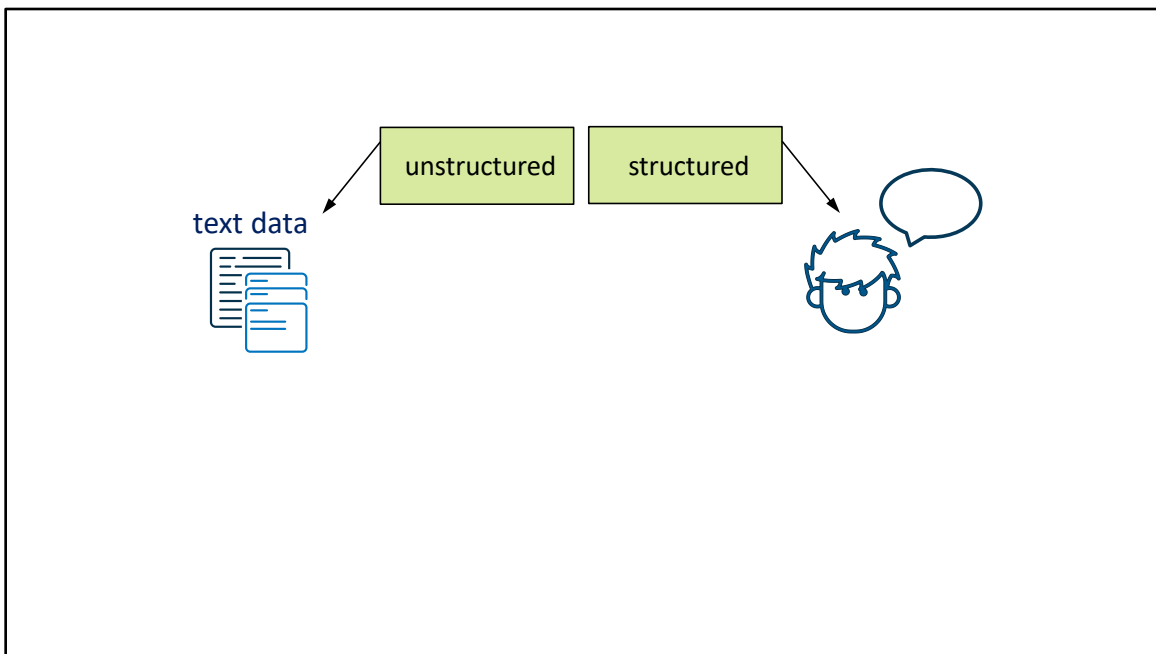
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Unstructured Text to Structured Data



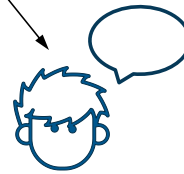
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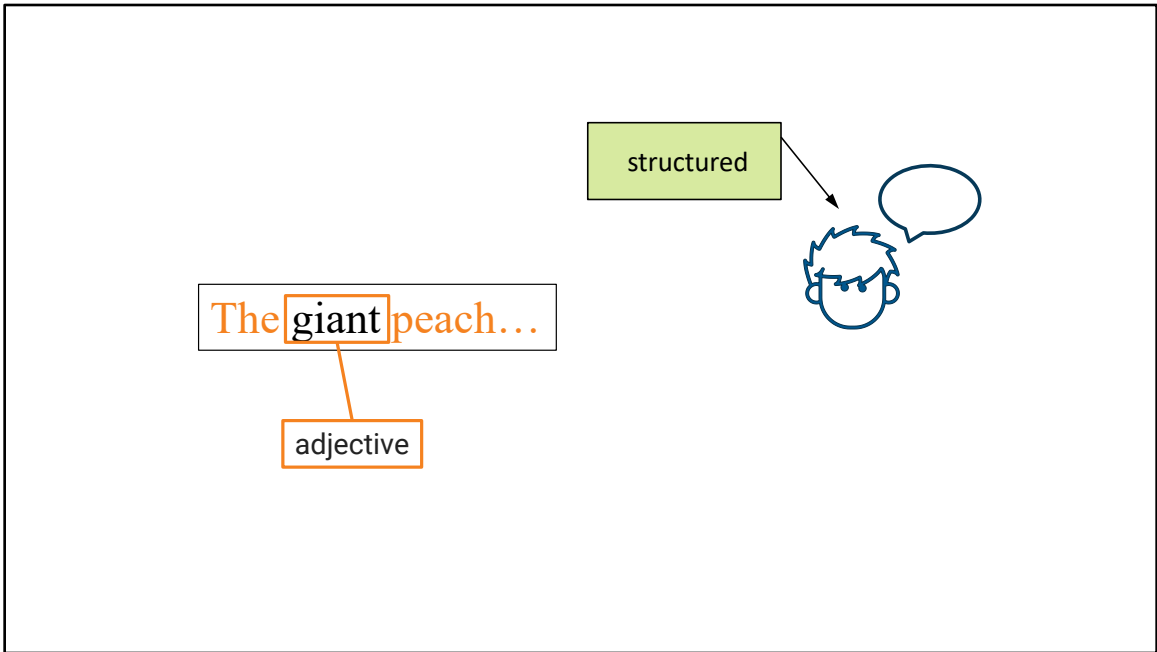
Text data are often described as unstructured data. Human language actually has a lot of structure.

The giant peach...

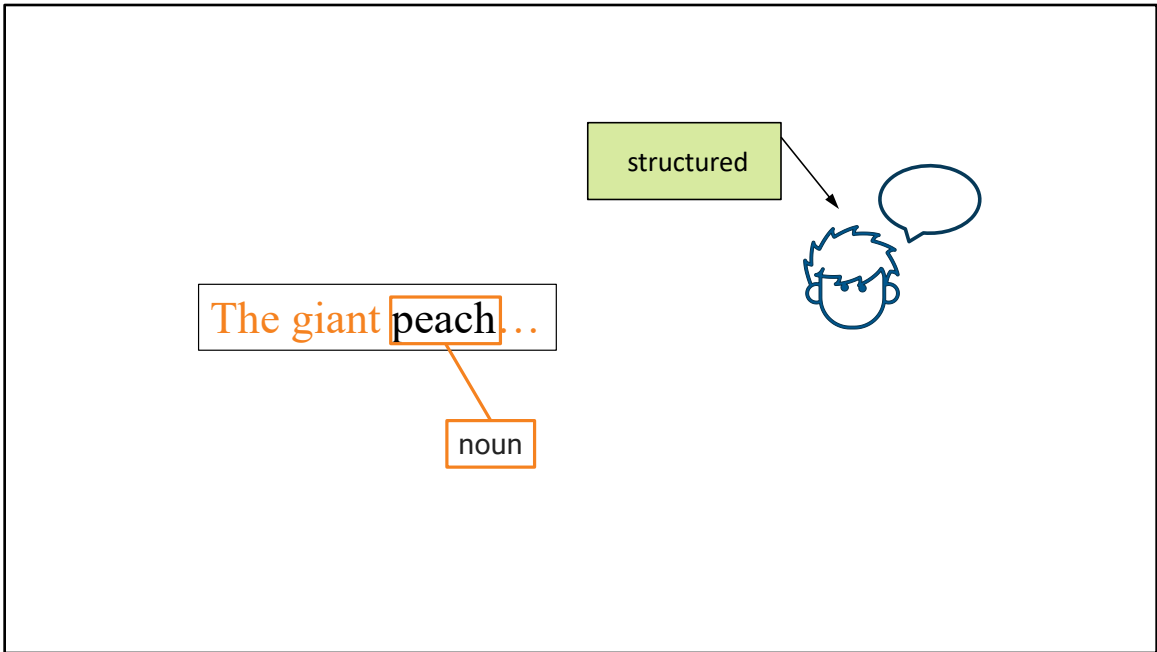
structured



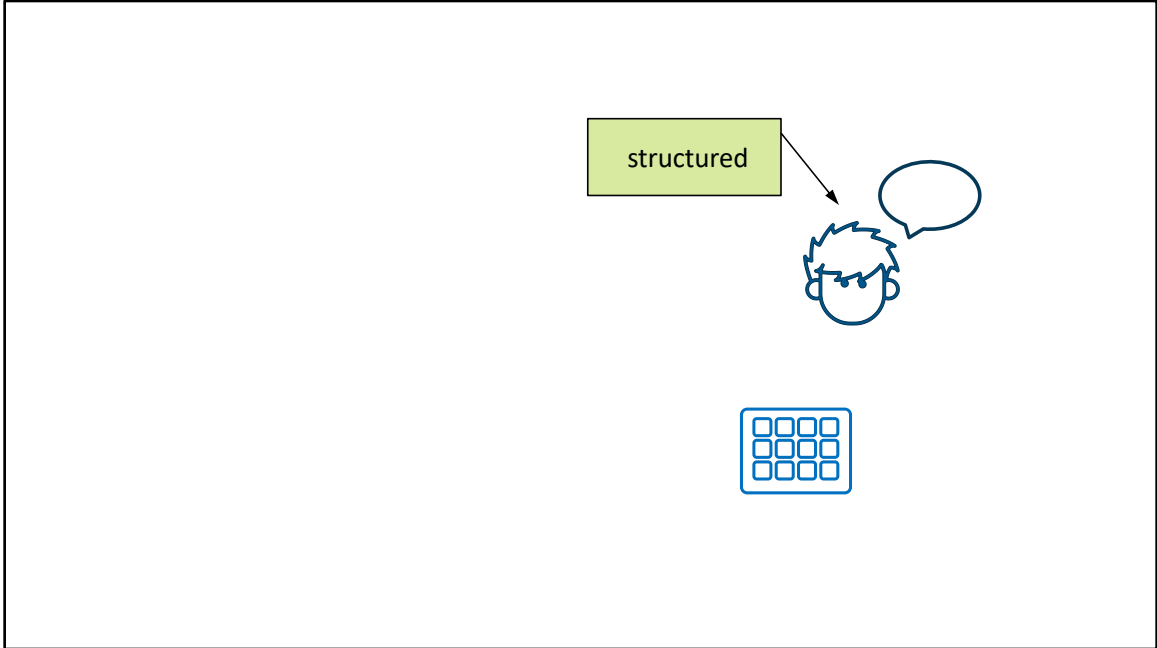
In the English language,



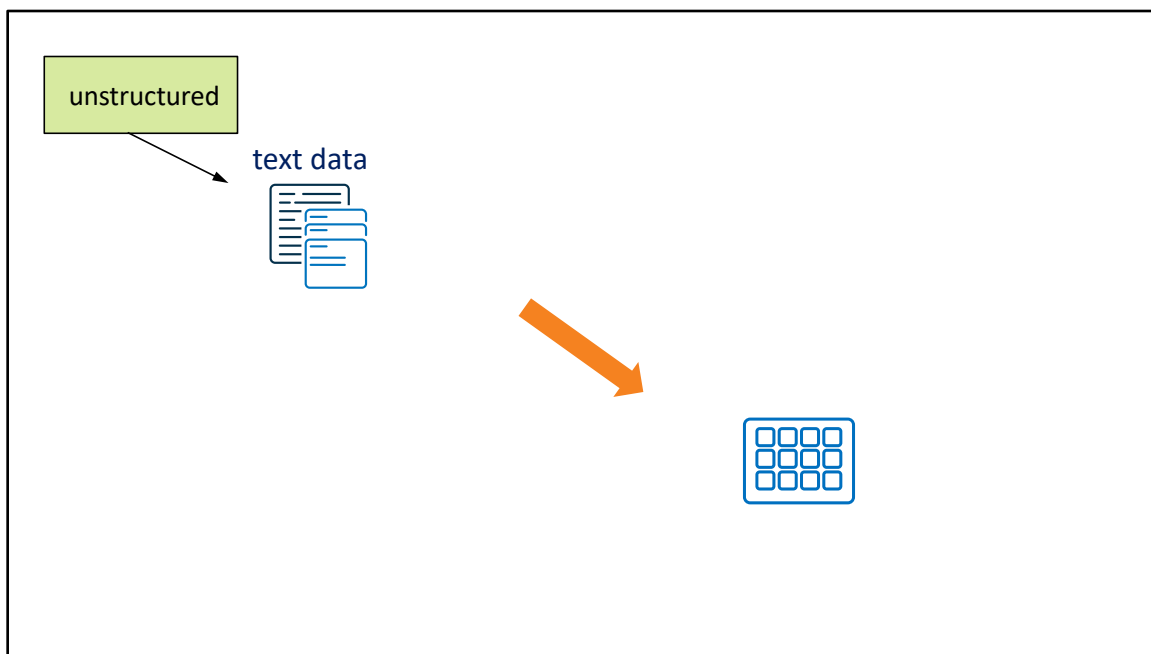
if you detect an adjective,



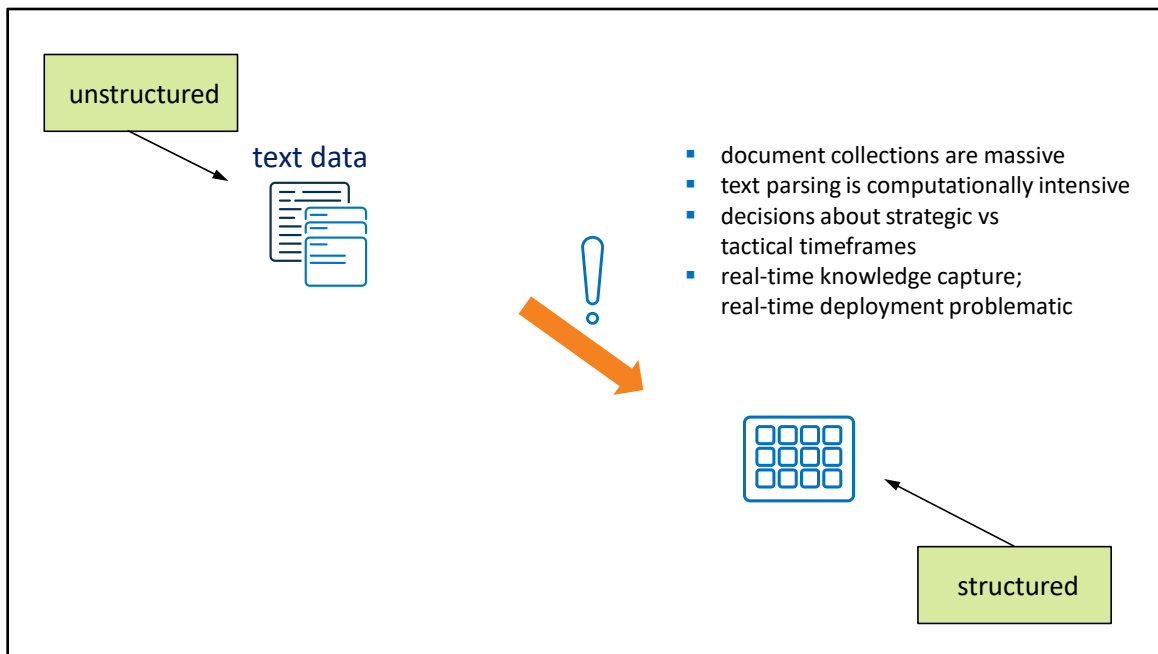
you can expect a noun to follow. In the Spanish language, an adjective comes after a noun.



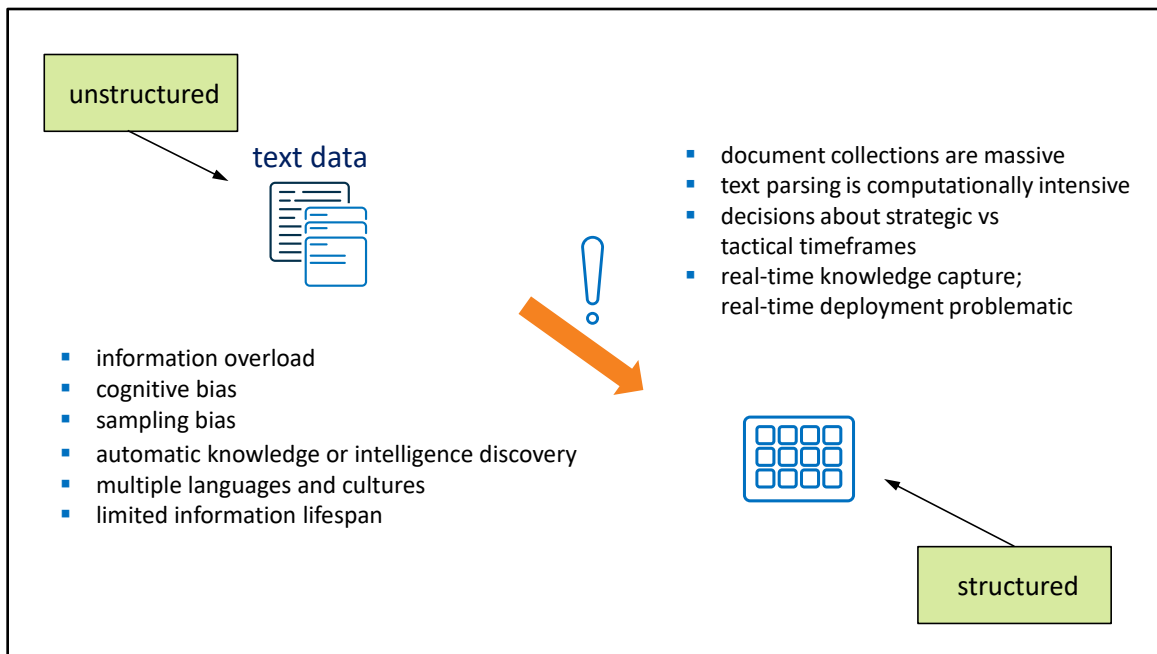
In the field of analytics, structure refers to a data set. Data sets are usually defined by simple row-column structures, or by hierarchical structures. Such data set structures are rigid and well-defined.



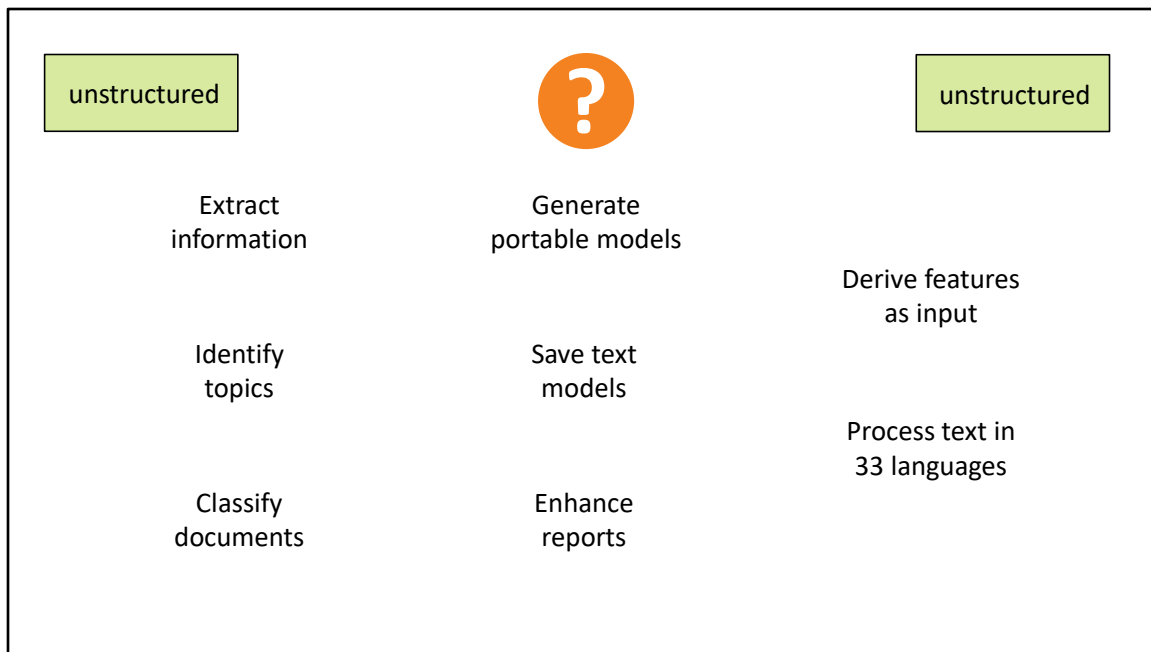
One goal of text analytics is to convert the weakly structured text data into rigidly structured analytics data.



There are many challenges to this conversion. Often document collections are massive. Text parsing is very computationally intensive. Consequently, processing times can be very lengthy. This leads to decisions about strategic versus tactical time frames. For data collection, you may require real-time knowledge capture. However, real-time deployment may be problematic. This is because conversions and translations are difficult to perform in real time.



The practical challenges to this conversion are information overload, cognitive bias, sampling bias (for example, social media), automatic knowledge or intelligence discovery, multiple languages and cultures, and limited information lifespan.



What can you do with unstructured and semi-structured text?

Extract information (including feature-level sentiment). Identify topics. Classify documents (including document-level sentiment). Generate portable models for contextual extraction, topics, categorization and sentiment. Save text models for deployment in batch, via API, in stream, or in Hadoop. Enhance business reports via native integration with data visualization. Derive new features as input for predictive modeling or artificial intelligence. Process text in 33 languages.

Unstructured Text to Structured Data: Practical Challenges

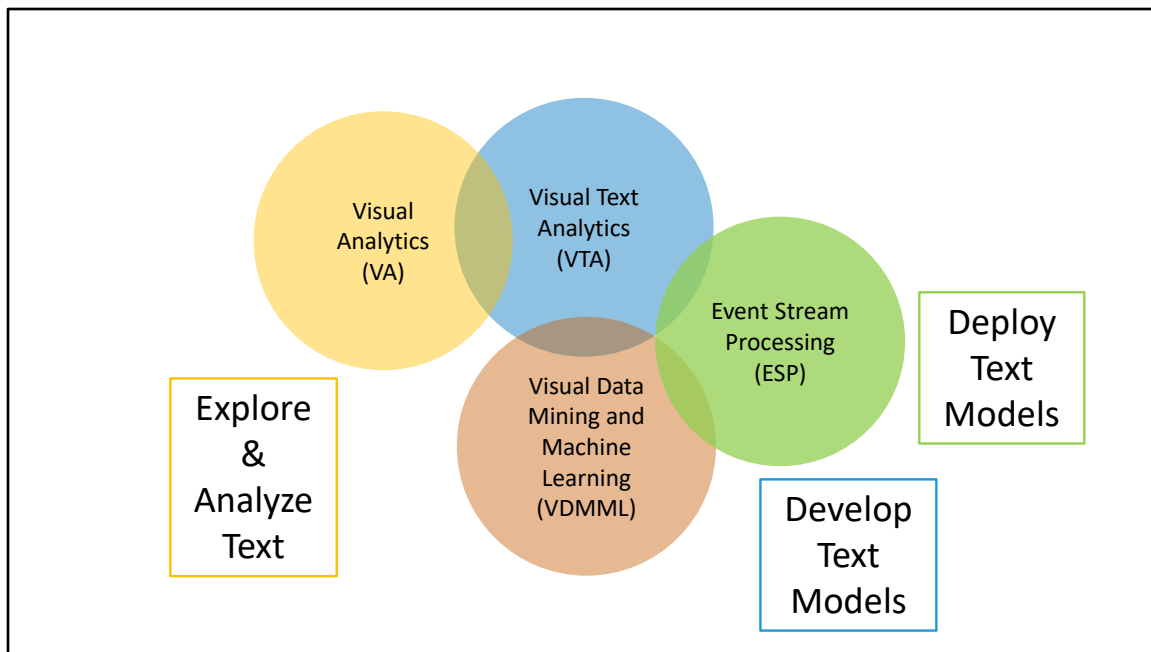
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SAS Products with Embedded Text Analytics Capabilities



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Text Analytics capabilities are found across multiple solutions at SAS, so how do you know which to use for a given situation?

Visual Text Analytics is our core solution for analyzing text, and contains the **most comprehensive** set of capabilities for exploring a document collection and developing text models for production.

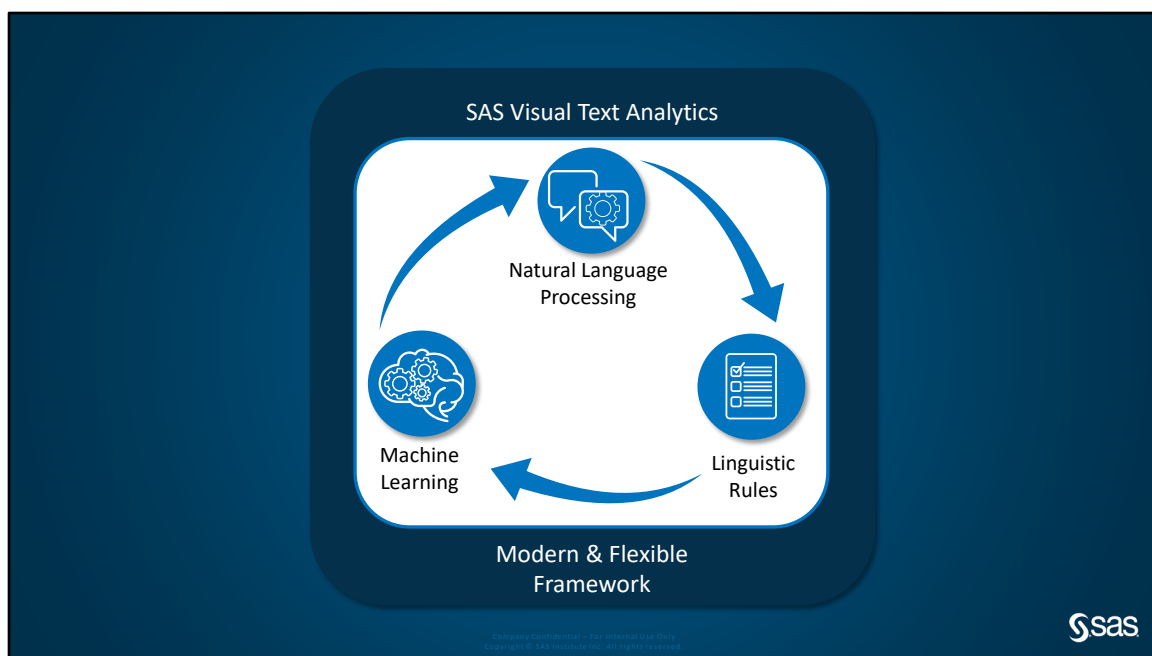
However, text capabilities have been embedded into other solutions in a limited fashion as well, and we'll touch on each of those intersections in a moment. The key thing to take away is that you can explore and analyze text collections in VA, but it's not intended for model creation...that is best left to VTA and VDMML. While VDMML contains only a subset of the text functionality that VTA does, it is well suited for incorporating text into other advanced analytics projects.

Finally, ESP can apply your text models to streaming data and thus enrich it, but ESP is not the interface where you would actually perform any text analytics itself.

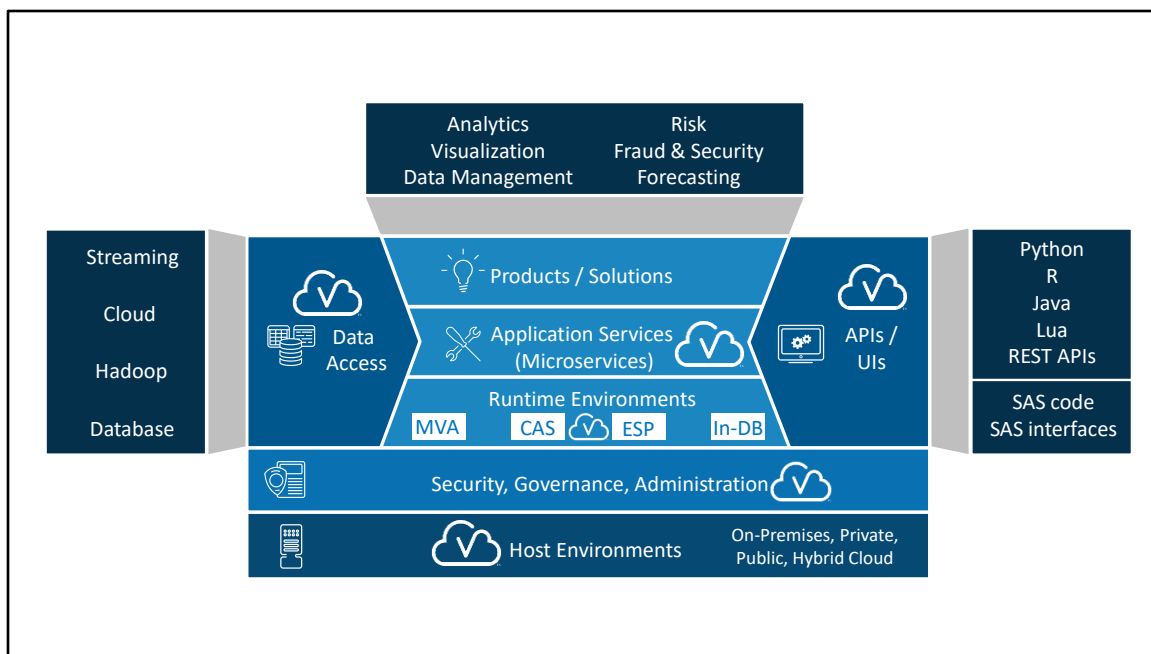
What Is SAS Visual Text Analytics?



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SAS Visual Text Analytics is the SAS offering designed to effectively extract insights from unstructured data in large scale. Offered on the SAS Viya architecture, Visual Text Analytics combines the power of natural language processing (NLP), machine learning (ML), and linguistic rules (algorithmically derived or human supplied).



Visual Text Analytics has both a Visual and Programming Interface. SAS solutions sit atop a stacked framework of architectural components built up including: common hosting environments, standardized implementations of security, governance, and administration, SAS runtime environments to process different kinds of analytical workloads, new SAS Viya software infrastructure to provide access to core SAS analytic engines, support for access to data from a range of sources, including standard DBMS, Hadoop, Cloud, and streaming (like IOT), support for client access, including SAS-built clients as well as open APIs for programming languages like R, Lua, Java, etc.

-

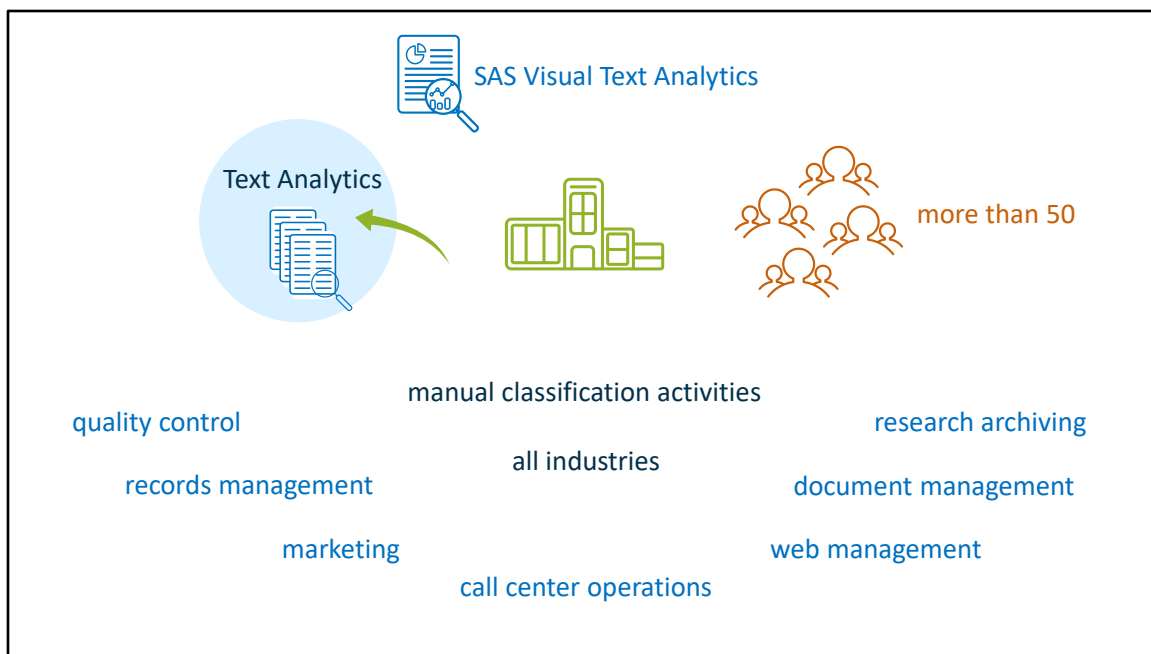
If you understand the text data well, then you could write linguistic rules to detect slang and sarcasm.

Stemming and part-of-speech tagging are relevant for many European languages. Stemming means detecting dictionary form and its variants, such as detecting singular and plural versions of a noun. Part-of-speech tagging identifies parts of speech such as verbs, nouns, and adjectives. Compound Decomposition is more relevant for languages like German, which accommodates combining words together to create a new word with a new meaning. Segmentation refers to breaking words into segments, for example when hyphens or slashes are used. Named entity recognition recognizes names of persons, organizations, locations, and products through predefined concepts. Semantic Parsing detects relationships between words in the sentence, for example, determining if a pronoun is the object or subject of a verb. Tokenization and sentence breaking identifies terms, sentences, and paragraphs based on punctuation and formatting.

Overview of SAS Visual Text Analytics



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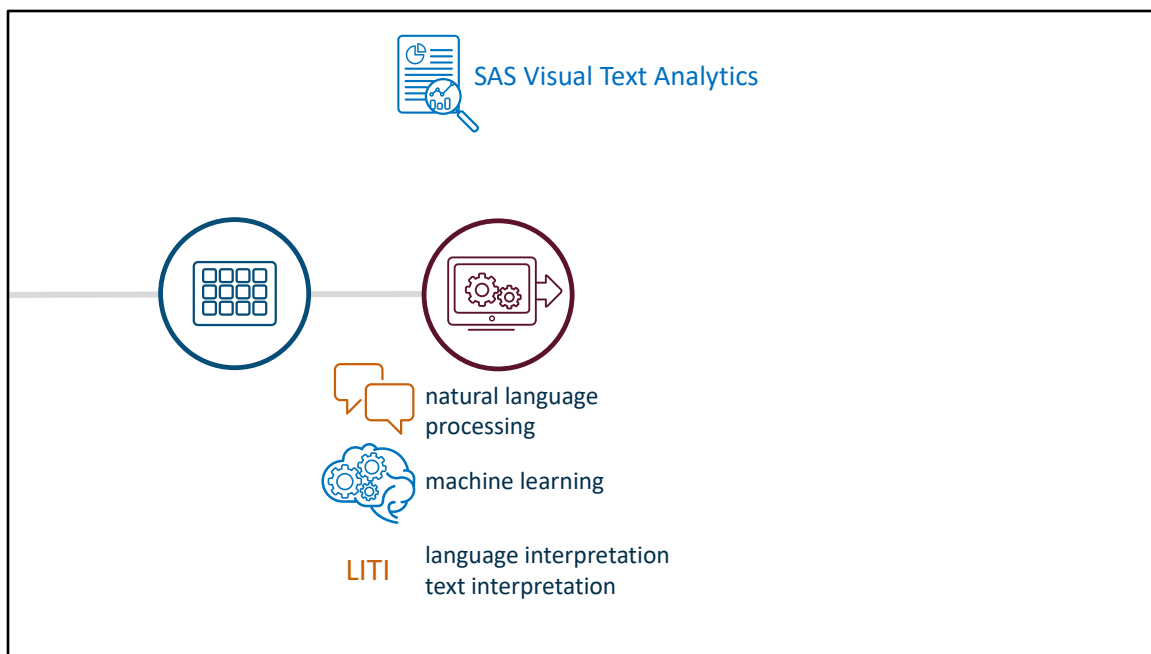
SAS Visual Text Analytics users come from enterprises that are often new to text analytics. These organizations often have more than 50 employees. Current operations include manual classification activities for large document collections. All industries are represented. Common business areas include quality control, records management, marketing, call center operations, research archiving, document management, web management, IT, or any other area that needs to classify documents or extract information.



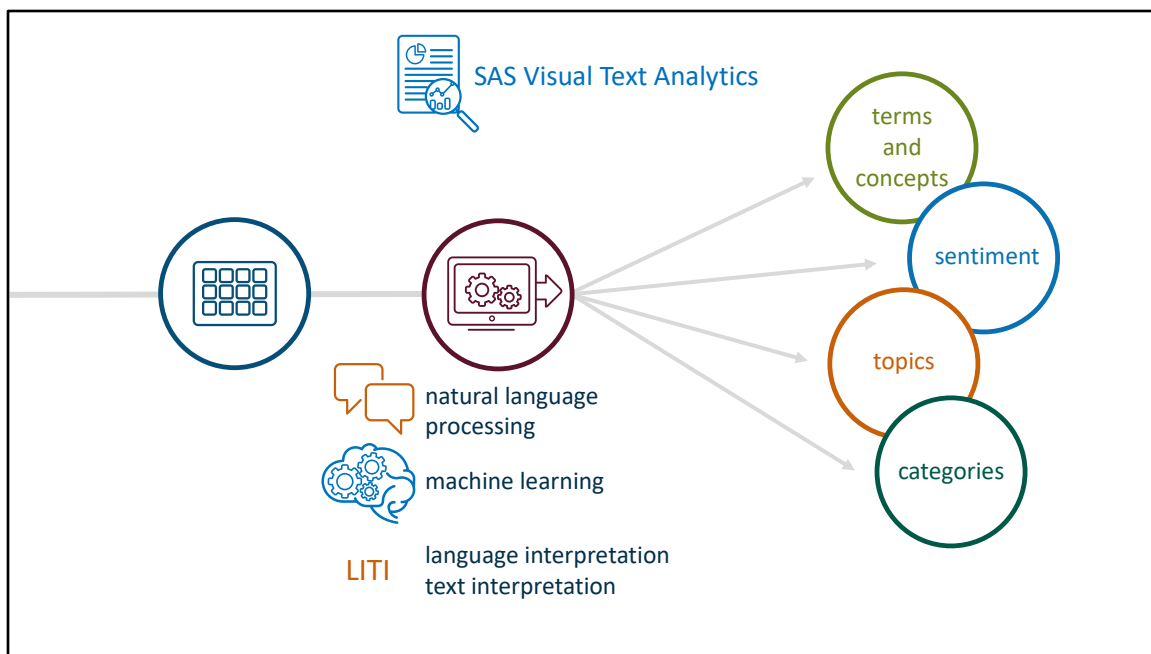
SAS Visual Text Analytics



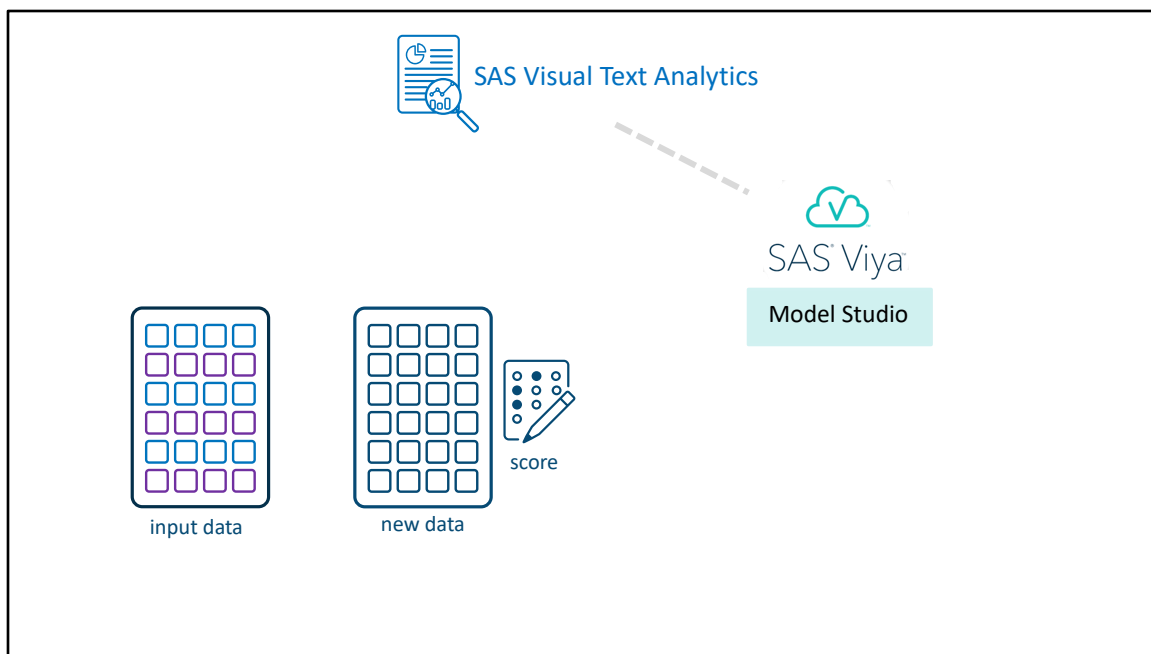
The SAS Visual Text Analytics process can be



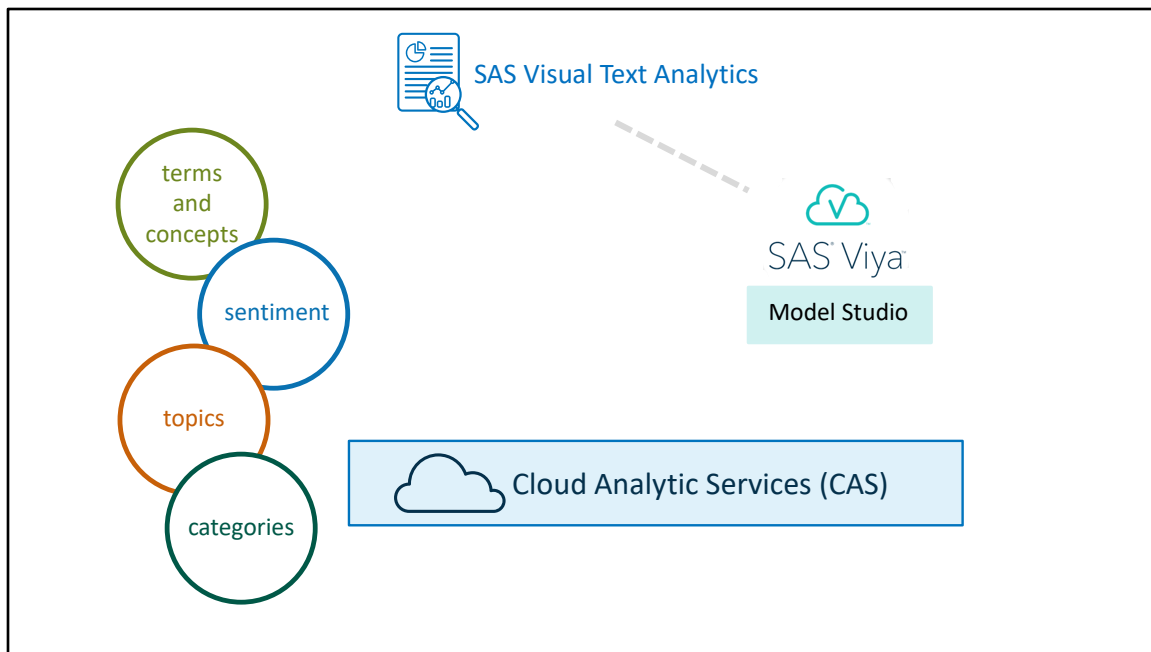
diagrammed as data being processed by the software using natural language processing, machine-learning, and the LITI (language interpretation for textual information) computing language. This process is represented



by a division of labor into four primary results: terms and concepts obtained from the Concepts node and the Text Parsing node, sentiment from the Sentiment node, topics from the Topics node, and categories from the Categories node.



When evaluating this as an enterprise solution, SAS Visual Text Analytics is fully integrated with SAS software. It's accessible via Model Studio in SAS Viya. SAS Visual Text Analytics can prepare input data and can score new data. And the interface makes it easy to manage multiple projects and to perform analyses on large or complex text data. You can include data from a SAS table, a CAS table, or documents converted into a CAS table using SAS Data Explorer.

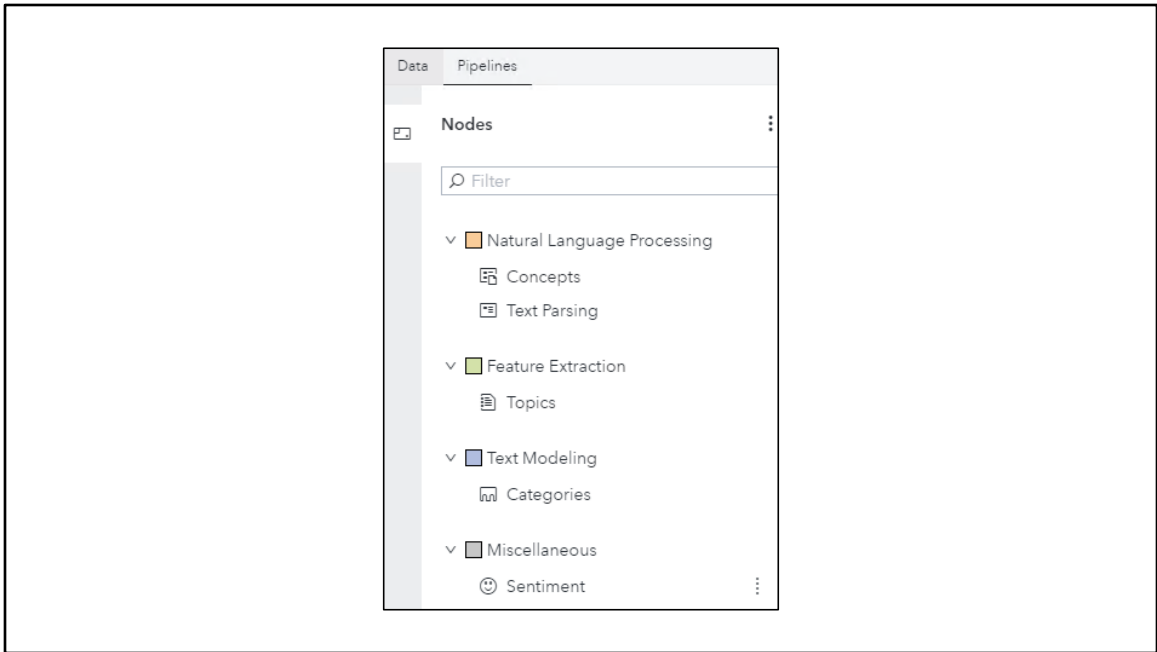


The analytic processing related to terms and concepts, sentiment, topics, and categories takes place on CAS tables in memory.

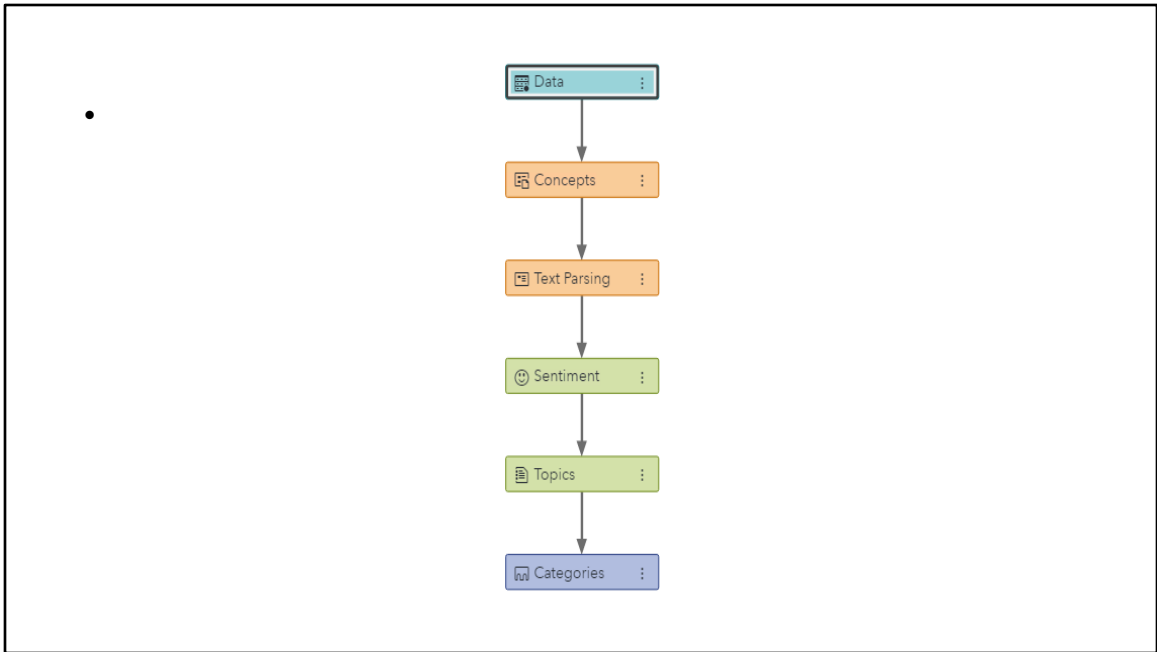
SAS Visual Text Analytics Pipeline



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SAS Visual Text Analytics pipelines have four major areas and five nodes. The four areas are Natural Language Processing, Feature Extraction, Text Modeling, and Miscellaneous, and the five nodes are Concepts, Text Parsing, Topics, Categories, and Sentiment.

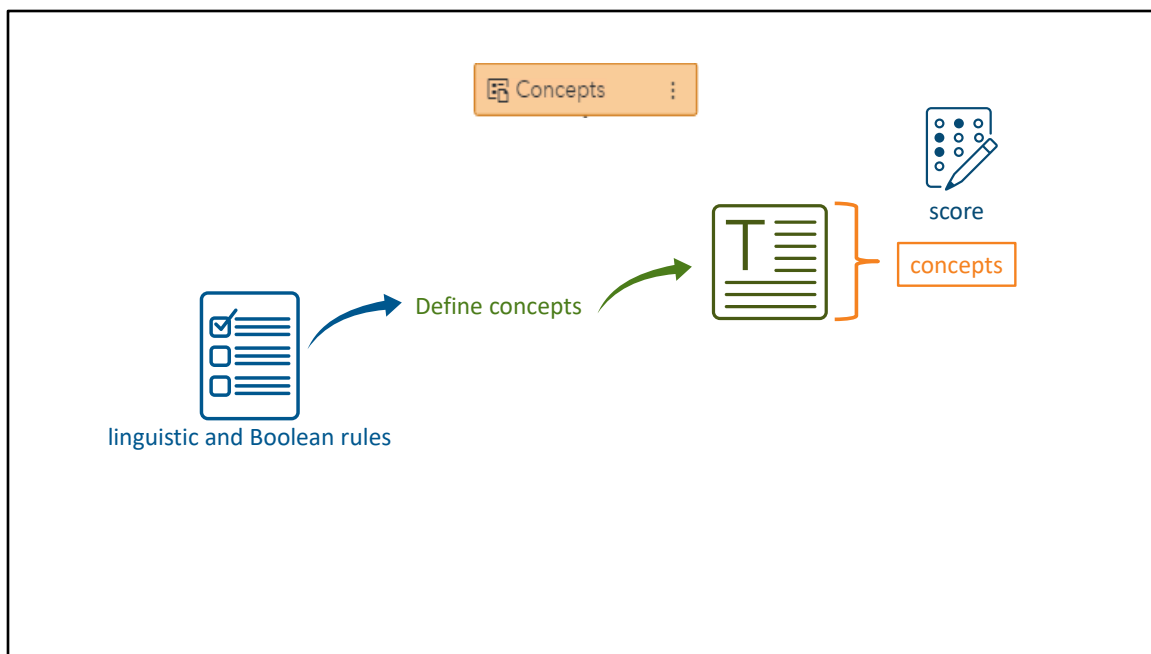


The default visual text analytics pipeline starts with the Data node and progresses through the five visual text analytics nodes. Note that the Concepts node appears before the Text Parsing node. This ensures that concepts will be used in topic derivation, category rule derivation, and sentiment analysis.

The Concepts Node: Concept Extraction



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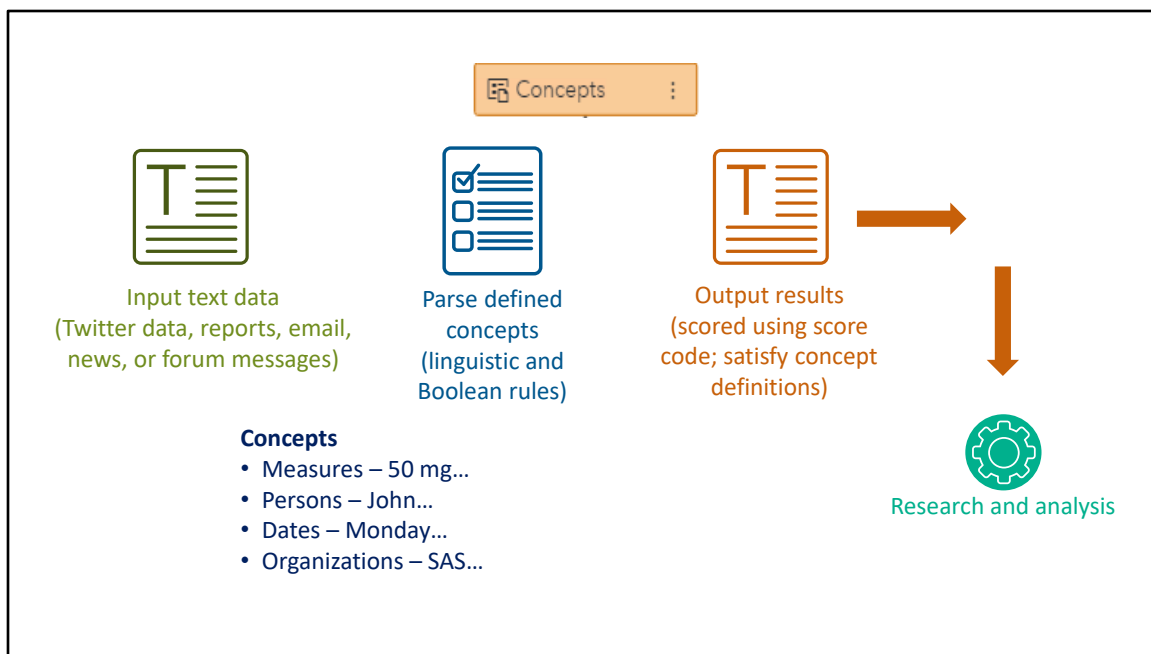
The Concepts node allows you to define concepts using linguistic and Boolean rules. Concepts are identified within the document, and a document can contain more than one concept. Concepts are extracted using the score code produced by the Concepts node.

The screenshot shows the SAS Concepts interface. At the top, there is a 'Concepts' button. Below it, a 'Custom Concepts (1)' folder is expanded, showing the '_Dosage' concept. A table defines the REGEX for this concept:

1	REGEX: [0-9]+[\.][0-9]+\s?mg\.
2	REGEX: [\d]+\s?mg\.

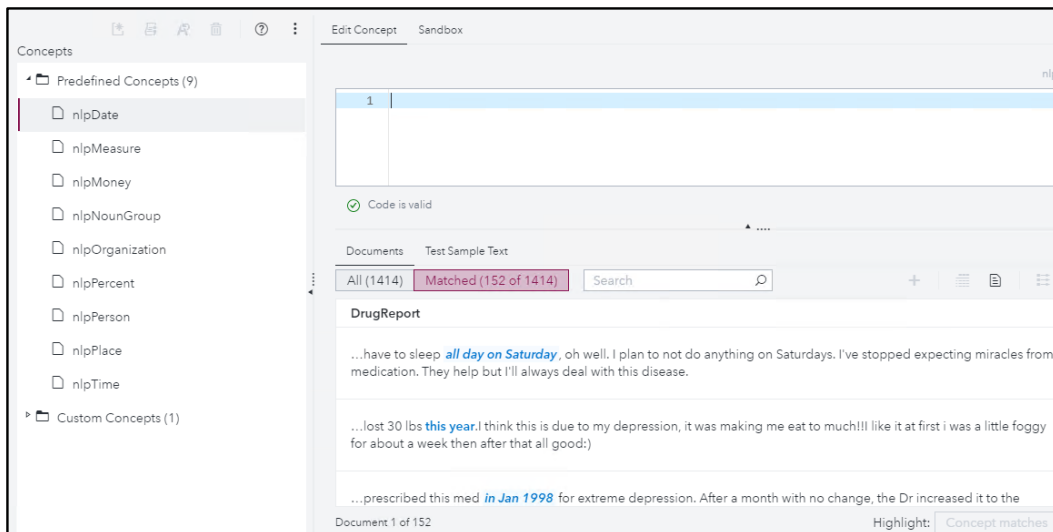
Below the table, a text box contains the example sentence: "I have been on 10mg prexifan for 6 months along with 2000mg noderall 1mg gemulex and fortifex when needed."

For example, the _DOSAGE concept is defined using regular expressions so that individual dosage values can be extracted from documents.

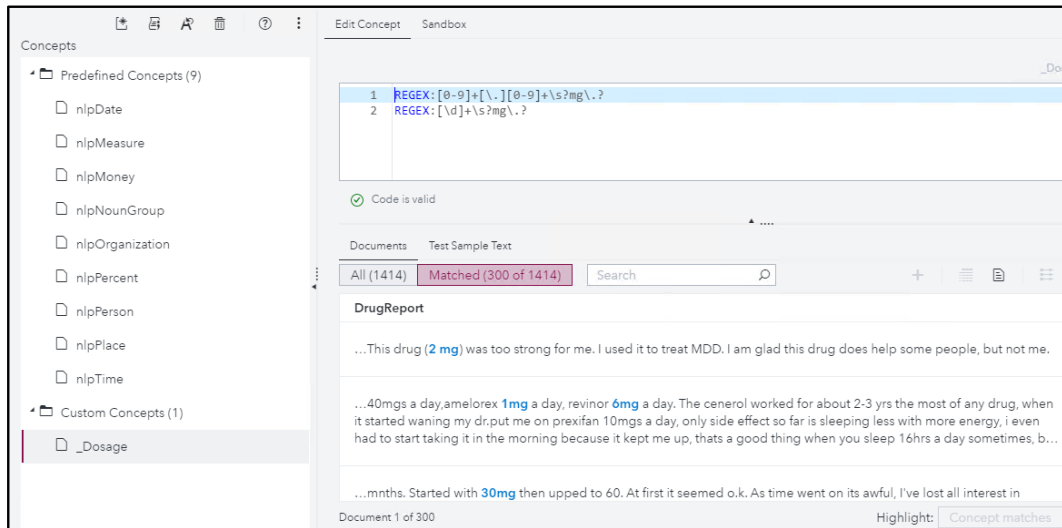


Concepts are the entities, facts, events, or key pieces of information in the input text data. The Concepts node parses, or extracts, concepts from the incoming documents by applying the concept definitions that you developed using linguistic and Boolean rules. Nine predefined concepts are supported by the Concepts node. These include concepts like measures, persons, dates, and organizations.

When a document is scored using the concept score code, the output contains the individual textual components that satisfy the concept definitions. Results are indexed, or fed, into existing systems for research and analysis.



If you request predefined concepts, and you open the Concepts node after it has run successfully, you will see the nine predefined concepts. Each predefined concept is named beginning with the letters NLP for natural language processing. When you select a predefined concept, for example, **nlpNounGroup**, you see the terms that satisfy the concept highlighted in the Documents pane under the Matched tab. You will not be able to see the source definition of the predefined concept in the Edit pane.

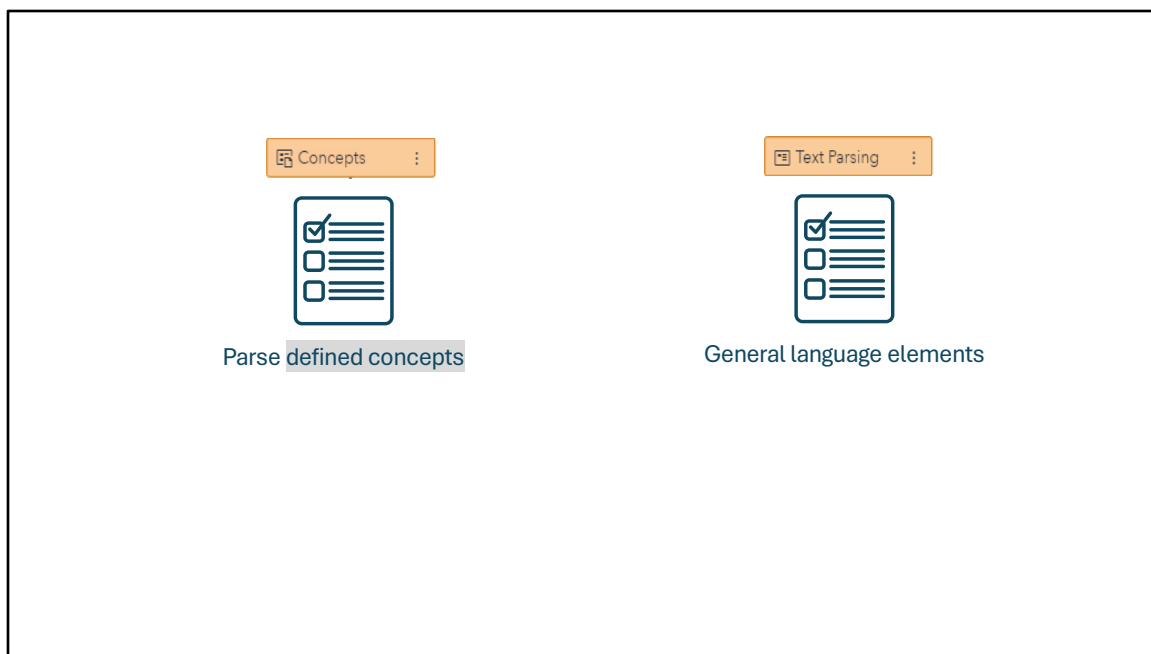


You can create a custom concept by supplying the rules for the concept in the Edit pane. The rules are taken from the LITI language. When you select the Matched tab in the Documents pane, you see highlighted concepts in the documents.

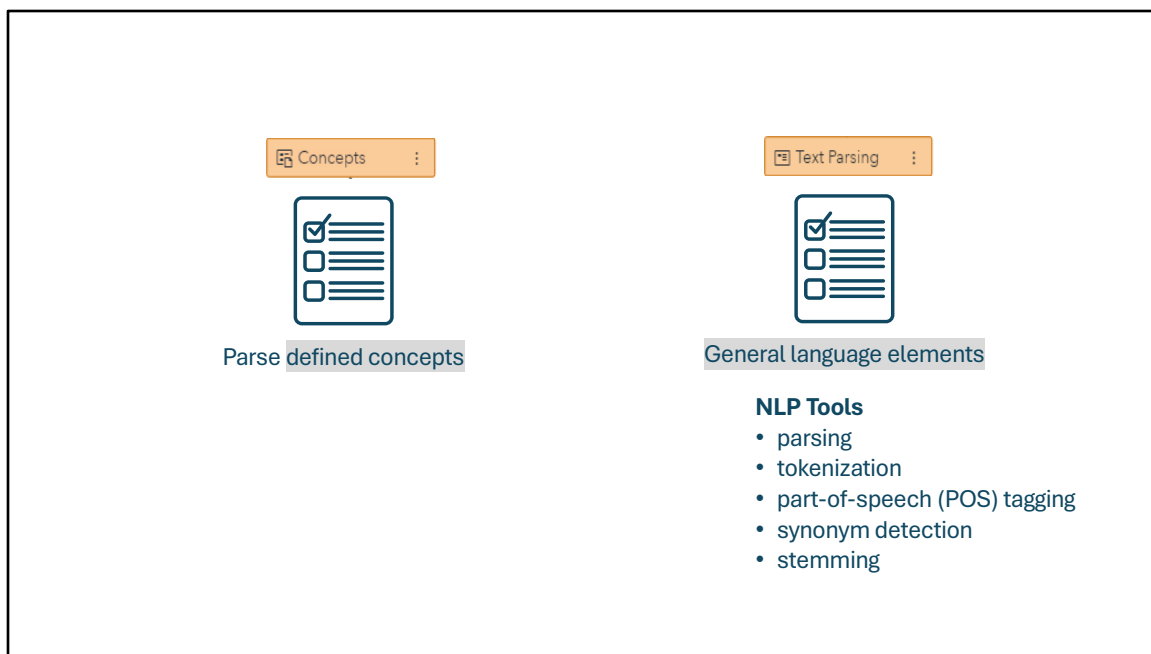
The Text Parsing Node: Term Extraction



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Although the Concepts node only parses for the defined concepts in the node,



the Text Parsing node parses for general language elements. The NLP tools supported by the Text Parsing node include parsing, tokenization, part of speech tagging, synonym detection, and stemming. In addition, if the Text Parsing node is preceded by the Concepts node, it will also parse with respect to the concepts, and any concepts found in the document collection will appear in the term table.

Kept Terms (1522)

Term	Role	Documents	Frequency
☐ not	ADV	658	1174
☒ take	V	677	1105
☐ depression	N	492	616
☐ feel	V	371	518
☐ year	N	395	502
☐ drug	N	341	485

Dropped Terms (6849)

Term	Role	Documents	Frequency
☐ i	PRO	1081	5003
☐ be	V	1129	4183
☐ have	V	1069	2849
☐ and	CONJ	1028	2746
☐ the	DET	904	2560
☐ to	PPOS	897	2386

Documents

All (1414) Matched

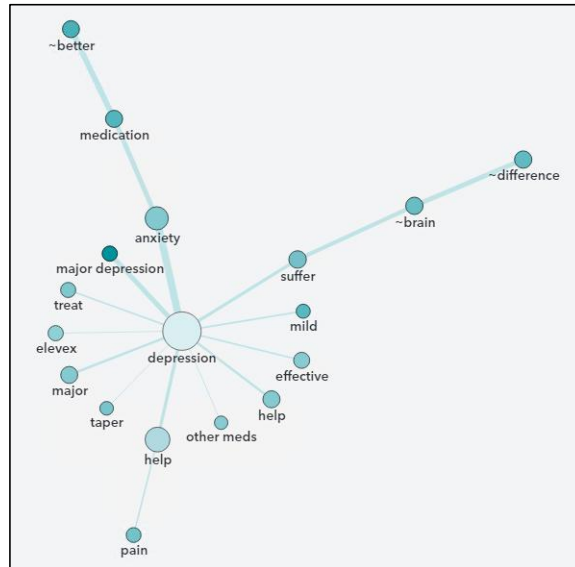
DrugReport

This medication made me gain 40 pounds it has been 2 years and I have only lost 10 pounds. Beware and watch your weight.

causing extreme anger, to the point that my family has become afraid of me. Doing things i would have considering doing before, like challanging the police, starting arguments, really, really wanting to beat the crap out of someone. Enough to the point that it scares me. But my doctor at this time refuses to take me off this medication. I am scared I will end up in jail for severly hurting someone.

Document 1 of 1414

When you open a Text Parsing node that has been run successfully, you see a Kept Terms pane, a Dropped Terms pane, and a Documents pane. You can select a term from either term table and find documents that contain the term by selecting the Matched tab in the Documents pane.



If you select a term in one of the tables, and click on the **Term Map** icon, you can see a term map with term associations related to the selected term. So, for example, if you select the term *depression*, associated terms such as *anxiety*, *help*, *treat*, and *suffer* are displayed.

The Topics, Categories, and Sentiments Nodes



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Topics (10)

☐	Topic	Created by	Docume... ↓
☒	+symptom, +withdrawal, off of, horrible, +dose	System	224
☐	+effect, +side effect, side, +side, sexual	System	223
☐	+work, ecstapin, best, +year, +try	System	223
☐	+day, +feel, mg, a day, +start	System	217
☐	abidal, +pain, just, +start, +work	System	208

Terms

All (1522) Matched Filter

☐	Term	Role	Documents	Frequency ↓
☐	not	ADV	658	1174
☐	take	V	677	1105
☐	depression	N	492	616
☐	feel	V	371	518
☐	year	N	395	502

Documents

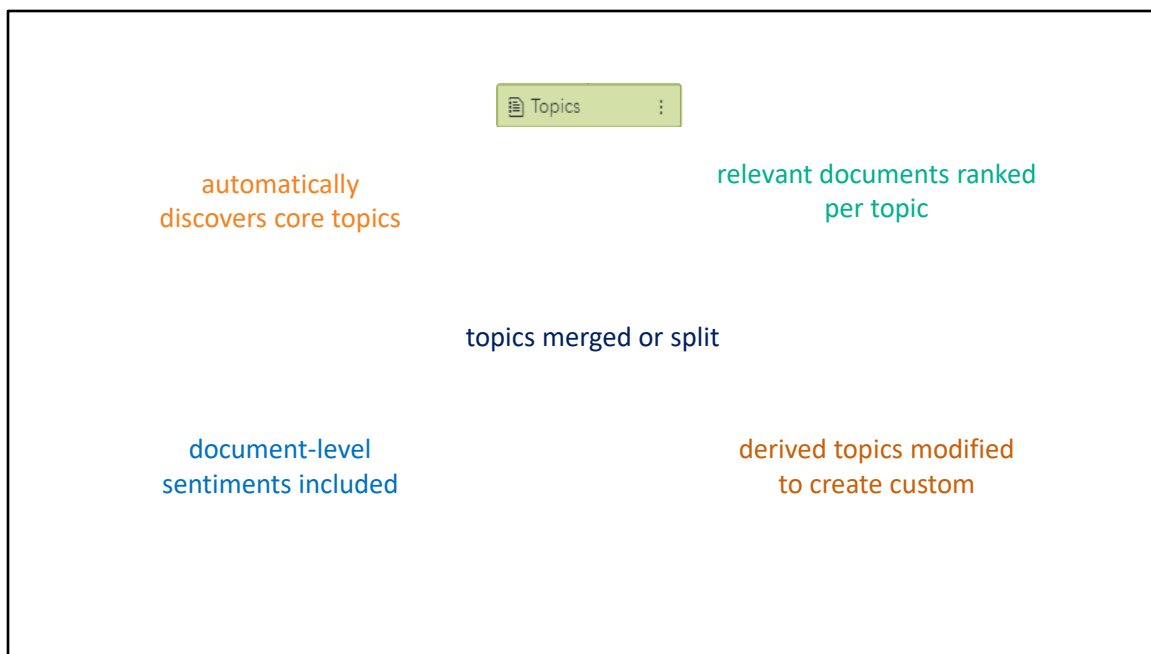
All (1414) Matched (224 of 1414) Search

DrugReport	Relevancy ↓	Sentiment
... taken ecstapin xr regularly for over 5 years . the side effects of going off of it (missing from one dose to several) are so severe i get anxiety just thinking about it. i recently went off of it suddenly and the withdrawal symptoms ranged from the most severe nausea i've ever experienced (including vomiting) to hyper-sensitivity through my fingers to a brain-clicking feeling like a cd that keeps skipping to anxiety and emotional upheav...	0.334	☹️
... Ecstapin for about 5 years . The withdrawal symptoms are horrible. I have brain zaps , nausea , vomiting , dizzy , can't concentrate, I fill like I am in a another world and vision problems I slowly took myself off of this deadly drug . I went from 150mg then to 37.5mg . I have been off it for about a week . My opinion with the withdrawals is like digging your own grave. I wouldn't recommend this drug to anybody.	0.330	☹️

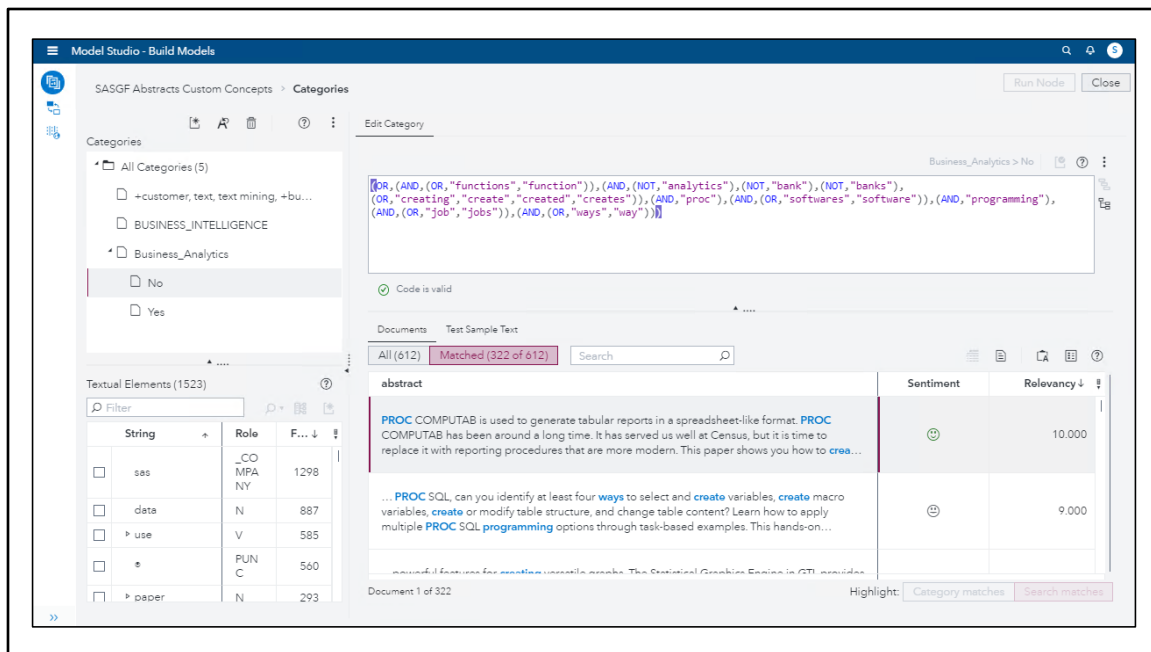
Document 1 of 224

Highlight: Topic matches Search matches

When you open the Topics node, you see a Topic pane, a Terms pane, and a Documents pane. You can select a topic, and then find matching terms and documents.



The Topics node automatically discovers core topics. The most relevant documents can be ranked with respect to a selected topic. Topics can be further merged or split. If a Sentiment node precedes the Topics node, document-level sentiments from the Sentiment node can be included. Derived topics can be modified to create custom topics.



When you open the Categories node, you see a Categories pane, an Edit window, a Textual Elements pane, and a Documents pane. If a variable in the input data has a role of category, the Categories pane shows the levels of the categorical variable, and if you select a level, you'll see the category rules associated with that level. You can also find the documents that match the categorical rules, and the terms that satisfy the category rules will be highlighted.

Categories

Model Studio - Build Models

SASGF Abstracts Custom Concepts > Categories

Categories

All Categories (5)

+customer, text, text mining, +bu...

BUSINESS_INTELLIGENCE

Business_Analytics

No

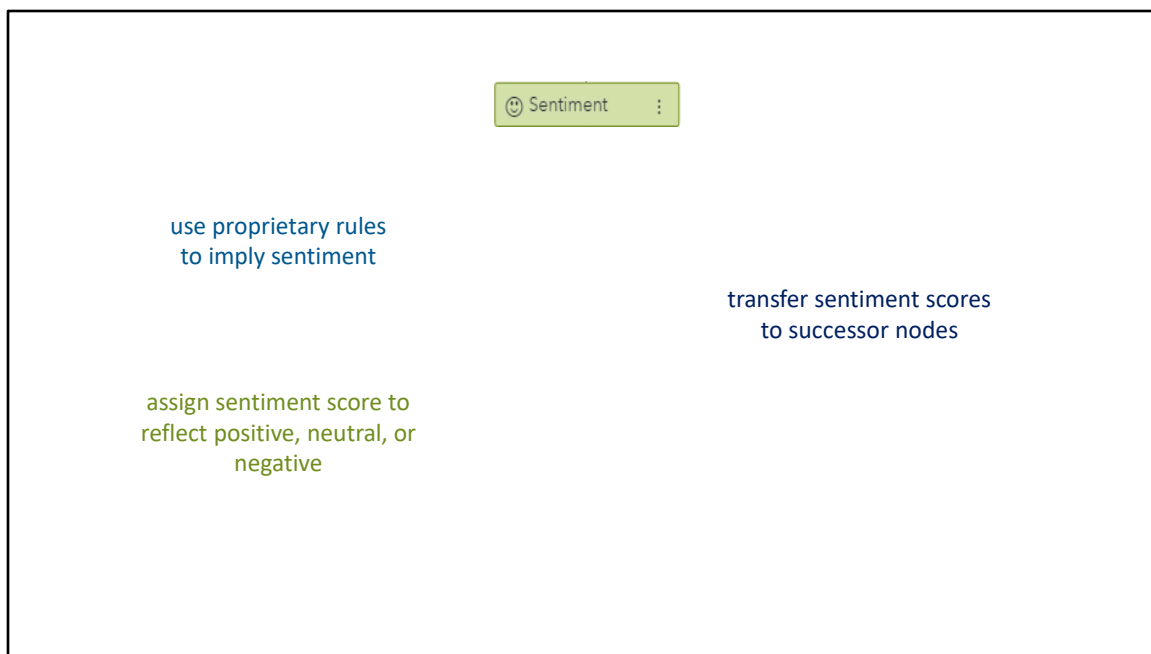
Yes

Boolean linguistic rules automatically derived

easy to understand

can be edited

When you supply a variable with the role of category, Boolean linguistic rules are automatically derived for that categorical variable. If you promote a topic to a category, then Boolean linguistic rules are also derived for the topic. The derived Boolean linguistic rules are easy to understand, and they can be edited. If you select results for the Categories node, you will see statistics on accuracy and specificity. Further customization or refinement by an analyst can help achieve the desired results.



The Sentiment analysis node uses proprietary rules that identify and analyze terms, phrases, and character strings to imply sentiment. Score code from the Sentiment node can be used to assign a sentiment score to each document to reflect positive, neutral, or negative sentiment. The sentiment scores can be transferred to successor nodes. Thus, the Topics node displays sentiment scores for each document if it is preceded by a Sentiment node.

Concepts, Terms, Topics, and Categories

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Question: SAS Visual Text Analytics Purpose



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Question: SAS Visual Text Analytics Purpose

What is the main purpose of using SAS Visual Text Analytics?
(Select all that apply.)

- a. concept extraction
- b. topic extraction
- c. content categorization
- d. sentiment extraction

Answer: SAS Visual Text Analytics Purpose

What is the main purpose of using SAS Visual Text Analytics?
(Select all that apply.)

- ☒ a. concept extraction
- ☒ b. topic extraction
- ☒ c. content categorization
- ☒ d. sentiment extraction

Correct answers: a, b, c, d

Question: SAS Visual Text Analytics Nodes



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Question: SAS Visual Text Analytics Nodes

Which of the following nodes can be used to extract information from within a document?

- a. Concepts
- b. Text Parsing
- c. Sentiment
- d. Topics
- e. Categories

Answer: SAS Visual Text Analytics Nodes

Which of the following nodes can be used to extract information from within a document?

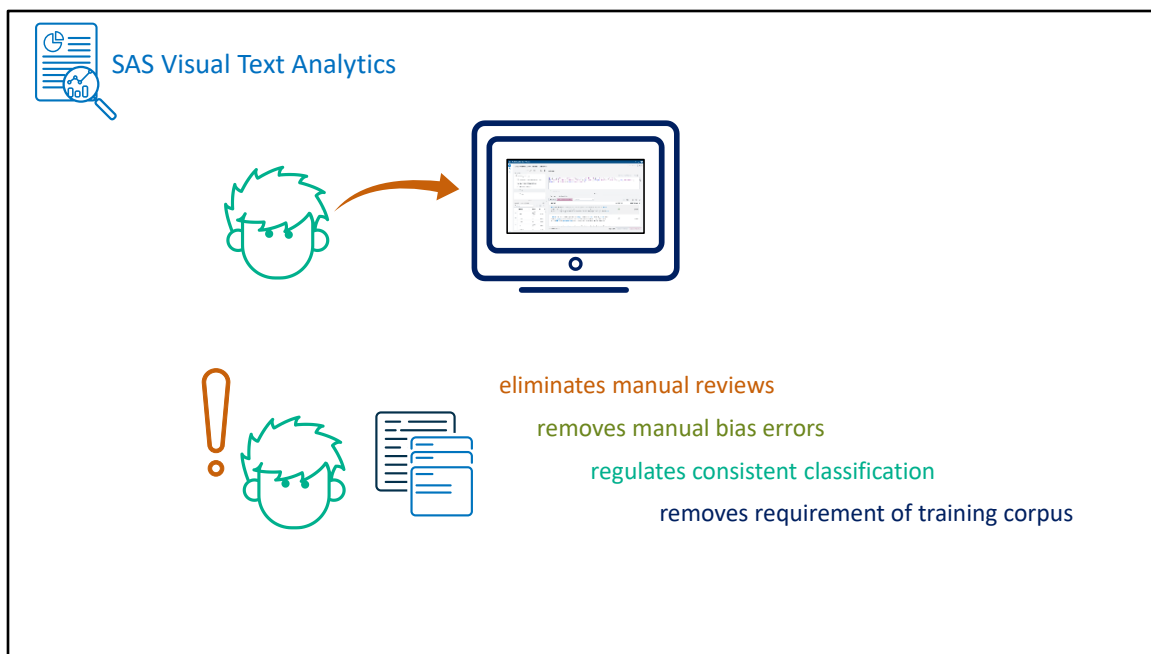
- ☒ a. Concepts
- b. Text Parsing
- c. Sentiment
- d. Topics
- e. Categories

a.

Benefits of SAS Visual Text Analytics



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There are many reasons to use SAS Visual Text Analytics for text analytics.

SAS Visual Text Analytics makes it possible for analysts to examine all their content, interrogate it, and define classification schemes in a single, simple-to-use GUI.

It eliminates laborious manual document reviews, removes manual bias errors, and regulates consistent content classification.

SAS Visual Text Analytics removes the requirement to create a training corpus in advance of defining classification rules.

Resource

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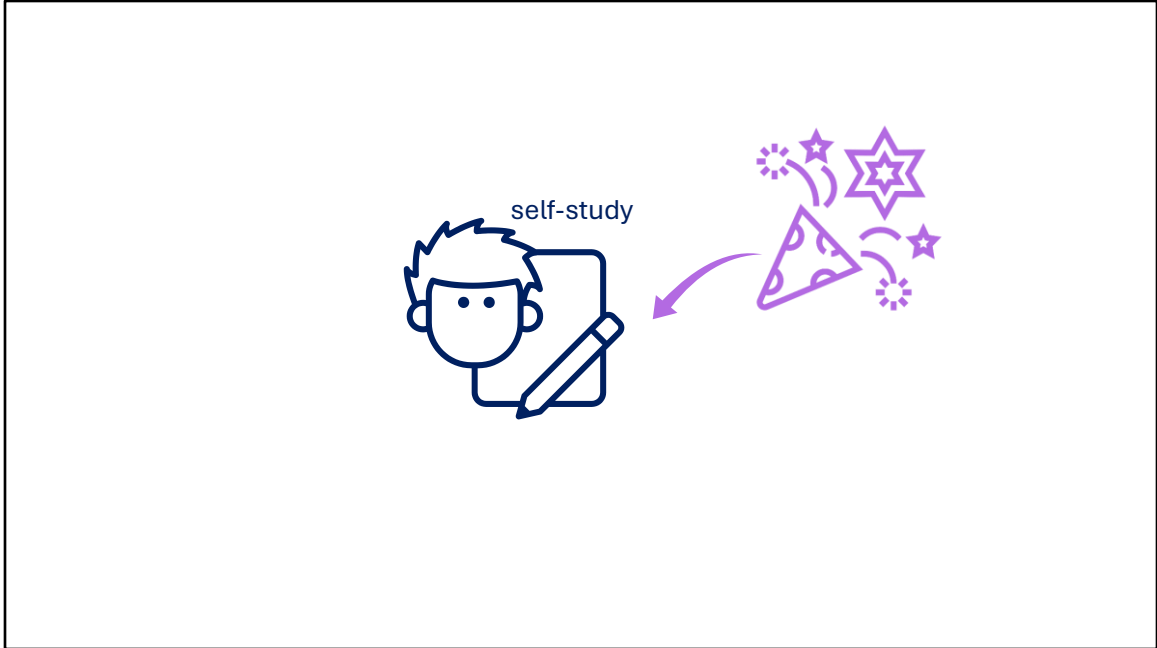
1.2 Language Challenges



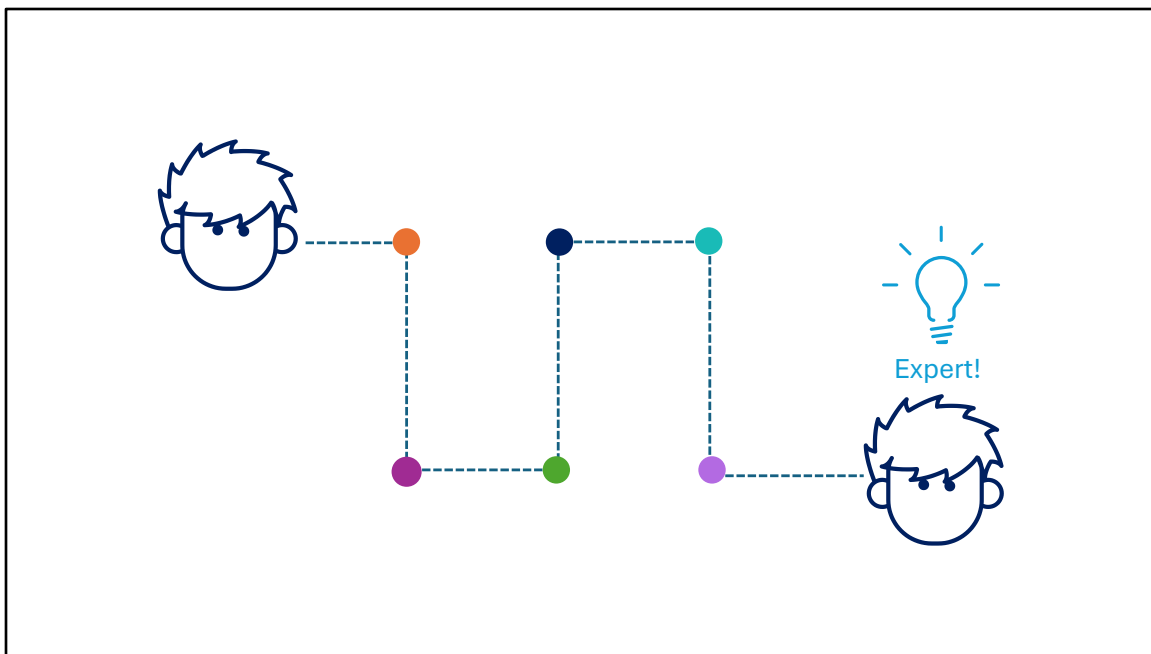
Self-Study Overview



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This section is classified as self-study because it is in the category of “fun facts that are not essential for success.” If you omit this section, it's unlikely that you will be less successful except for relatively rare cases.



However, if you are on a path to become a text analytics expert, some information in this section can equip you to tackle more serious document conversion and data transcoding problems.

Self-Study

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.

